Malawi University of Science and Technology

Malawi Institute of Technology

Discrete Mathematics

MATH 2101

Module Writer

Madalitso Dorothy Kanyika

Module Reviewer

Date

June 2023

TABLE OF CONTENTS

LIST OF SYMBOLS

$\sim$	Relation
$\iff$	If and only If
$\exists$	There exist
$\in$	Member of
$\notin$	Not a member of
$\cup$	Union
$\cap$	Intersection
$\mid$	Condition on
$\to$	Mapped to

unit 1

ALGEBRAIC STRUCTURE

Introduction

In this unit, we are going to look at the Algebraic structures such as Cross product, Relations and equivalence classes, binary compositions and its properties. We are also to discuss about different types of groups such as Semigroup, Monoid , Abelian and their properties, Permutations groups ,Subgroup ,Cyclic groups as well as looking at the Introduction to rings and Fields. We will also introduce some operations that will help us to determine the relationship between variables.

Intended Learning Outcomes

By the end of this Unit, you should be able to:

a. Describe cross products of sets

b. Describe relations and Equivalence classes

c. Differentiate among semigroup , monoid and group

d. Differentiate between a ring and a field

e. Define a subgroup

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:
Cross product, relations , equivalence classes, binary composition , semigroup, monoid , groups, Abelian group, permutation, sub-group, cyclic groups, rings and fields.

1.1 THE CROSS PRODUCT

Reflection : When you reflect the word Cross product what comes in your mind? What does it produce? How do you calcu – late it and where does the formula comes from?

1.1.1 Ordered Pairs

If we have a pair of elements that occur in a particular order and enclosed with brackets it means we are dealing with a set of ordered pairs.Now we are going to look on the formal definition of an ordered pair.

Definition 1.1. *If a , b are two elements then a set of an ordered pair is the represented as (a , b) where a is the first component and b is a second component.*

If the position of the component is changed then the ordered pair is changed.

$i.e \ \ (a, b) \neq (b, a)$

Definition 1.2 (Equality Ordered Pairs)**.** *Let (u , v) and (x , y) be sets of ordered pairs. The two ordered pairs are equal i.e (u,v)= (x,y) only if u=x and v=y.*

Example: Find the value of $(2x - 2, \ \ y - 1) = (x + 3\ ,\ 4)$.

Solution: By equality of ordered pairs the two ordered pairs will be equal only if the corresponding first components and second components are equal. So

$$\begin{aligned} 2x - 2 &= x + 3; \ \ y - 1 = 4 \\ 2x - x &= 3 + 2; \ \ y = 4 + 1 \\ x &= 5\ ; \quad y = 5 \end{aligned}$$

Therefore the values of x, y are 5 , 5

Example: What is the value of $x, \ y$ if the ordered pairs $(x, \ y)$ and (3, 5) are equal?

Solution: Since $(x, y) = (3, 5)$ then by definition we have $x = 3$ and $y = 5$.

1.1.2 Cartesian (cross) Product of two sets

Cross products are also known as Cartesian product. Let us consider A and B to be two non-empty sets and a Cartesian product $(A \times B)$ is defined as a set of all ordered pairs (a, b) where $a \in A$ and $b \in B$. If $A = B$ then $A \times B$ is called the Cartesian square of set A represented as A^2. That is $A^2 = \{(a, b)\ a \in A$ and $b \in A\ \}$.

Example: If $A = \{3, 4, 5\}$ and $B =$ {1 , 2}.Find the value of $A \times B$, $B \times A$, B^2

Solution: Given A = { 3, 4 , 5 } and { 1, 2 }. Then the Cartesian product:

$$\begin{aligned} A \times B &= \{3,4,5\} \times \{1,2\} \\ &= \{(3,1),(3,2),(4,1),(4,2),(5,1),(5,2)\} \end{aligned}$$

In the same way we can find the Cartesian product of $B \times A$ as follows:

$$\begin{aligned} B \times A &= \{1,2\} \times \{3,4,5\} \\ &= \{(1,3),(1,4),(1,5),(2,3),(2,4),(2,5)\} \end{aligned}$$

Since if we have $A = A$ then $A \times A$ is a set of Cartesian product of set A represented by A^2. Therefore the Cartesian product of B^2 from the example above, is given as follows:

$$\begin{aligned} B^2 &= \{1,2\} \times \{1,2\} \\ &= \{(1,1),(1,2),(2,1),(2,2)\} \end{aligned}$$

From the first two cases of the example above we have seen that $A \times B$ and $B \times A$ do not have the same ordered pairs. Therefore $A \times B \neq B \times A$.

Example: If $A = \{x, y, z\}$ and $B = \{y, z\}$. How many sets of ordered pairs are there in the Cartesian product of $A \times B$

Solution: Using $n(s)$ as the number of elements in set S we have

$$\begin{aligned} n(B \times A) &= n(A) \times n(B) \\ &= 3 \times 2 \\ &= 6 \end{aligned}$$

Another way of getting a solution above is by first coming up with the Cartesian product of $A \times B$ and finally count the number of ordered pairs.

1.1.3 Relations and its types

A Relation defines the relationship between two different set of information.

Definition 1.3. *Let A and B be sets. A binary relation or simply a Relation from A to B is a subset of* $A \times B$*.*

Definition 1.4. *Suppose R is a relation from* A *to* B *such that* $a \in A$ *and* $b \in B$ *then the following hold:*

i. If $(a, b) \in R$ *, we say "a is R- Related to b " written as a R b*

ii. $(a, b) \notin R$ *we then say "a is not R-related to b written as "* $a \not R\, b$*.*

If R is a relation from a set A to itself i.e $A^2 = A \times A$ then R is a relation on A. The domain of a relation R is the set of all first element of the ordered pairs which belong to R and the Range is the set of second element in R.

Example: Let $A = (1, 2, 3)$ and $B = (x, y, z)$ and let $R = \{(1, y), (1, z), (3, y)\}$. Then we say R is a relation from A to B since R is a subset of $A \times B$. With respect to this relation

$$1Ry\ ,\ 1Rz\ ,\ 3Ry\ \text{ but }\ 1 \not R\, x,\ 2 \not R\, y\ 2, \not R\, z$$

$$\text{The domain of } R = \{1, 3\}$$

$$\text{The range of } R = \{y, z\}$$

Example: How many relations are there on a set with n elements?

Solution A is a subset of $A \times A$. Because $A \times A$ has n^2 elements when A has n elements , and a set with m elements has 2^m subsets. Therefore there are 2^{n^2} subsets of $A \times A$. Thus, we have 2^{n^2} relation on a set with n elements.i.e there are $2^{3^2} = 2^9 = 512$ relation on the set $\{a,\ b,\ c\}$

Note: In $A \times B$, If A has x elements and B has y elements such that m represents xy elements. Then a set with m elements has 2^m subsets.
In Discrete Mathematics, the relation can be of various types and we will differentiate based on connections between sets.

a. **Empty Relation**

A relation is said to be empty if the element of the set have no relation with each other. $R = \emptyset$

b. **Universal Relation**

In this relation , every element of set A has a relation with every element of set B. For example if $A = \{1, 2, 3\}$ and $B = \{a, b\}$ then $R = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$ is a universal relation.

c. **Identity Relation**

In this relation , every element of A has a relation to itself only. It is denoted by $I = \{(a, a), a \in A\}$

Example: Suppose $A = \{a, b, c\}$, then identity Relation is given by $I = \{(a, a), (b, b), (c, c)\}$

Example: For the set $A = \{\mathbf{1}, \mathbf{2}, \mathbf{3}\}$, the identity relation is given by $I = \{(1, 1), (2, 2), (3, 3)\}$

d. **Inverse Relation**

Suppose that there are two sets A and B such that they have a relation from

A to B. i.e $R \in A \times B$. Then we define an inverse relation as a relation where the elements position of each pair in $A \times B$ are interchanged by swapping their position. Thus if $R = (a, b)$ then $R^{-1} = (b, a)$

Example: Given that relation $R = \{(u, v), (w, x)\}$. Then the inverse relation of R denoted by $R^{-1} = \{(v, u), (x, w)\}$.
Example: Given that relation $R = \{(1, 2), (2, 3)\}$. Then the inverse relation of R denoted by $R^{-1} = \{(2, 1), (3, 2)\}$.

d. **Reflexive Relation**

A relation R on a set A is called reflexive if $(a, a) \in R$ for every $a \in A$
Example: If $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (2, 2), (1, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$. Is the relation reflexive?
Solution: By definition the relation is reflexive as for every $a \in A$ $(a, a) \in R$. Thus (1, 1),(2, 2), (3 , 3),(4 ,4)$\in R$.
Example: Consider the following relation on $A = \{1, 2, 3, 4\}$:

$$
\begin{aligned}
R_1 &= \{(1,1)(1,2),(2,1),(2,2),(3,4),(4,1),(4,4)\} \\
R_2 &= \{(1,1)(1,2)(2,1)\} \\
R_3 &= \{(1,1)(1,2),(1,4),(2,1),(2,2),(3,3),(4,1)(4,4)\} \\
R_4 &= \{(2,1)(3,1),(3,2),(4,1),(4,2),(4,3)\} \\
R_5 &= \{(1,1)(1,2),(1,3),(1,4),(2,2),(2,3),(2,4)(3,3)(3,4)(4,4)\} \\
R_6 &= \{((3,4)\}
\end{aligned}
$$

Which of these relations are reflexive?
Solution: The relation R_3 and R_5 are reflexive because they both contain all pairs of the form (a, a) namely (1 , 1) , (2 , 2) , (3 , 3) and (4 , 4). The other relations are not reflexive because they do not contain all these ordered pairs. i.e R_1, R_2, R_4 and R_6 are not reflexive because (3 , 3) is not in any of these relation.

e. **Symmetric Relation**

A relation is symmetric if $(a, b) \in R$ is true then $(b, a) \in R$ is true for all a , b$\in A$. Symmetry relation is denoted by $aRb \Longrightarrow bRa$ $\forall$ a , b$\in A$.

Example: Given that $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (1, 2), (2, 1), (2, 3),$ $(3, 2)\}$ is the relation R symmetric or not?

Solution $=$ By definition a relation is symmetric as every $(a, b) \in R$ we have $(b, a) \in R$ .Therefore we have $(1, 2)(2, 1)(3, 2)(2, 3) \in R$ as symmetric relation.

f. **Transitive Relation**

Suppose there are three elements in a set A i.e. a, b and c. The relation R on a set A is transitive relation if "a " has a relation with "b " and "b " has a relation with "c " then "a " will also have a relation with "c ". Transitive relation is denoted as follows: if $(a, b) \in R$, $(b, c) \in R$ then $(a, c) \in R \ \forall \ a, \ b, \ c \in A$

Example: Given that $A = \{1, \ 2, \ 3\}$. Then the relation $R = \{$(1 , 2) (2 , 3) (1 , 3)$\}$ is transitive.

Example: Consider the following relation on $A = \{1, 2, 3, 4\}$:

$$
\begin{aligned}
R_1 &= \{(1,1)(1,2),(2,3),(1,3),(4,4)\} \\
R_2 &= \{(1,1)(1,2)(2,1)(2,2)(3,3)(4,4)\} \\
R_3 &= \{(1,3)\ (2,1)\} \\
R_4 &= \emptyset, \text{the empty relation} \\
R_5 &= A \times A
\end{aligned}
$$

Which of these relations are transitive?

Solution The relation R_3 is not transitive since (2 , 1) (1 ,3) $\in R_3$ but $(2, 3) \notin R_3$. All other relations are transitive.

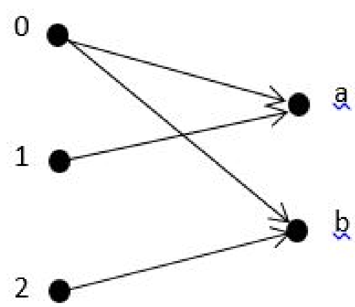
g. **Equivalence relation**

A relation is equivalence if it is reflexive , Symmetric and Transitive. **Example**: Show that a set $A = \{1, 2, 3\}$ is an equivalence relation.

Solution: Since $A = \{1, 2, 3\}$ then the relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2),$ $(2, 1), (2, 3), (3, 2), (1, 3), (3, 1), \}$. From the relation we have $\{(1, 1), (2, 2), (3, 3)\}$ is reflexive on R , **(1, 2),** $\{(2, 1), (2, 3), (3, 2)\}$ is symmetry on R and $\{(1, 2), (2, 3)(1, 3)\}$ is transitive on R. Therefore the Set A on relation R is an equivalence relation.

We can also represent relations using graphs or arrows. For instance , if $A = \{0,\ 1, 2\}$ and $A = \{a,\ b\}$ such that the relation from A to B is $\{(0,\ 1)\ (0,\ b)\ (1,\ a)\ (2,\ b)\}$. The graphical or arrow representation of the relation is given as follows:

Arrow-diagram



Graph

R	a	b
0	1	1
1	1	0
2	0	1

1.1.4 Equivalence Classes

Definition 1.5 (Equivalence Classes / Partitions)**.** *Suppose R is an equivalence relation on a set S. For each $a \in S$, Let [a] denote the set of elements of S to which “a ” is related under R ; that is [a]= {x | (a, x) ∈ R}. We call [a] the equivalence class of a in S; any b ∈ [a] is called a representative of the equivalence class.*

Definition 1.6. *The quotient set of S by R denoted by S / R is the collection of all equivalence classes of elements S under an equivalence relation R given by* $\{[a] \mid a \in S\}$

Example: Consider the relation R= {(1, 1), (1,2) ,(2,1), (2,2) ,(3,3) } on S= {1, 2 ,3}. We note that [1]=[2]= {1 , 2} and [3] = {3}. Therefore S/R= {[1] , [3]} is a partition of S. Therefore we can choose either {1 , 3} or {2 ,3} as a set of representatives of the equivalences classes.

Example: Let R be the following equivalence relation on the set A= { (1, 1), (1,5) ,(2,2), (2,3) ,(2,6) (3, 2), (3,3) ,(3,6), (4,4) ,(5,1) (5, 5), (6,2) ,(6,3), (6,6) }. Find the equivalence classes of R

Solution: To find the equivalence classes of R we need to find the partition of A induced by R. Therefore we have the following:

i. Those elements related to 1 are 1 and 5 hence [1] = {1,5}

ii. Next we pick an element that does not belong to [1] say, 2. Those elements related to 2 are 2 ,3 and 6 , hence [2]= {2 ,3 ,6}

iii. The only element which does not belong to [1] or [2] is 4. So the only element related to 4 is 4. Thus [4]= {4}

Therefore the partition of A induced by R= [{1,5}, {2 ,3 ,6} , {4}]

1.1.5 Binary Compositions

Definition 1.7. *Let R be a relation from a set A to a set B and S is a relation consisting of ordered pairs (a, c), where $a \in A$, $c \in C$ and for which $\exists$ an element $b \in B$ such that $(a , b) \in R$ and $(b , c) \in S$. The composition (or composite) of R and S is denoted* *by R o S*

Example: What is the composite relations of R and S , where R is a relation from {1 ,2 ,3} to {1, 2, 3 ,4} with $R = \{(1,1),(1,4),(2,3),(3,1),(3,4)\}$ and S is a relation from { 1 ,2 , 3 , 4} to { 0, 1, 2} with $\{(1,0),(2,0),(3,1),(3,2),(4,1)\}$

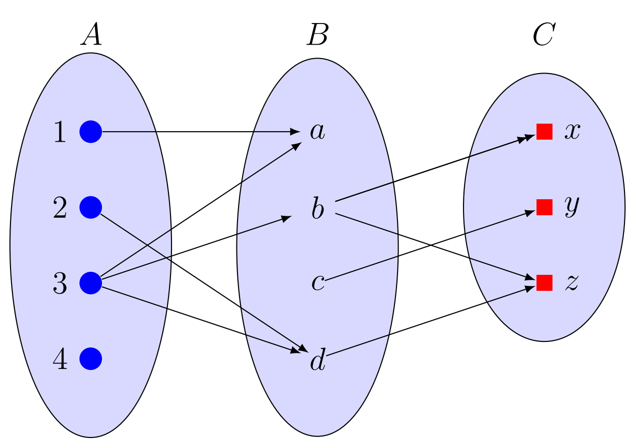
Solution

1 (R o S) 0	since 1 R 1	and 1 S 0
2 (R o S) 1	since 2 R 3	and 3 S 1
2 (R o S) 2	since 2 R 3	and 3 S 2
3 (R o S) 0	since 3 R 1	and 1 S 0
1 (R o S) 1	since 1 R 4	and 4 S 1
3 (R o S) 1	since 3 R 4	and 4 S 1

Therefore $R\ o\ S$ = {(1 ,0) (2, 1) (2, 2) (1, 1) (3, 0) (3 ,1) } .

Example: Let A= {1, 2 ,3, 4} , B= {a, b ,c, d} and C= {x ,y, z}. And let R = $\{(1,\ a),\ (2,\ d),\ (3,\ a),\ (3,\ b),\ (3,\ d)\}$ and S= $\{(b,\ x)\ ,(b,\ z),(c\ ,y)\ ,(d\ ,z)\}$. What is the composite relation of R and S

Solution



$$2\ (R \circ S)\ z \quad \text{since } 2\ R\ d \quad \text{and } d\ S\ z$$
$$3\ (R \circ S)\ x \quad \text{since } 3\ R\ b \quad \text{and } b\ S\ x$$
$$3\ (R \circ S)\ z \quad \text{since } 3\ R\ d \quad \text{and } d\ S\ z$$

Therefore $R\ o\ S = \{(2,\ z), (3,\ x),\ (3,\ z)\}$.

Activity 1a

1. Find the number of relation from $A = \{a,\ b,\ c\}$ to $B = \{1\ ,2\}$.

2. If $(4a\ ,4) = (3a + 2\ ,b - 1)$. Find the values of $a,\ b$

3. Consider the following five relation on the set $A = \{1,\ 2,\ 3\}$:

   R= $\{(1\ ,1),\ (1\ ,2),\ (1,\ 3),\ (3\ ,3)\}$

   S= $\{(1\ ,1),\ (1\ ,2),\ (2,\ 1),\ (2\ ,2), (3\ ,3)\}$

   T= $\{(1\ ,1),\ (1\ ,2),\ (2,\ 2),\ (2\ ,3)\}$

   $\varnothing$= empty relation

   A $\times$ A= Universal relation

   Determine whether or not each of the above relation on A is

   (a) Reflexive

   (b) Symmetric

   (c) Transitive

4 If A= $\{4,\ 5,\ 6\}$ and A= $\{7,\ 8\}$. Find the cartesian product of $A \times B$

1.2 Groups

1.2.1 Operation , Semigroup and Monoid

We start by defining an operation with its properties that will be used through out this section.

Definition 1.8 (Operation). *Let S be a non empty set. An operation on S is a function* $*$ *from* $S \times S$. *The set* S *and an operation* $*$ *on* S *is denoted by (S ,* $*$ *)*

Definition 1.9 (Closure Property). *Suppose S is a set with an operation* $*$ *, and suppose* A *is a subset of* S. *Then* A *is said to be closed under* $*$ *if* $a * b$ *belongs to* A *for any element a and* $b \in A$.

Example: Consider the set N of positive integers. Addition (+) and multiplication ($\times$) are operations on N.
Example: Let A denote a set of even positive integers. Then A is closed under Addition and Multiplication since the sum and product of any even numbers are even.
Example: Let B denote a set of odd positive integers. Then B is closed under Multiplication but not addition since the sum of odd numbers are even $e.g$ 3+5=8.

Definition 1.10 (Associative law). *An operation* $*$ *on a set* S *is said to be associative or to satisfy the associative law , if for any a , b , c in* S *we have*

$$(a * b) * c = a * (b * c).$$

Definition 1.11 (Commutative law). *An operation* $*$ *on a set* S *is said to be commutative or to satisfy the commutative law , if for any a , b in* S *we have*

$$(a * b) = (b * a).$$

Definition 1.12 (Identity element). *Consider an operation $*$ on a set S. An element e in S is called an identity element for $*$ if for any element a in S, we have*

$$a * e = e * a = a.$$

An element e is called a left identity or a right identity when $a*$ e=a or e $*$ a=a where a is any element in S

Definition 1.13 (Inverses). *Suppose an operation $*$ on a set S does have an identity element e. The inverse of an element a in S is an element b such that, we have*

$$a * b = b * a = e.$$

Note: 0 has no multiplacative inverse.

Definition 1.14 (Cancellation law). *An operation $*$ on a set S is said to satisfy the left cancellation law or the right cancellation law, as according to*

$$(a * b) = (a * c) \text{ implies } b=c \text{ or}$$
$$(b * a) = (c * a) \text{ implies } b=c.$$

Definition 1.15 (Semigroup). *Let S be a nonempty set with an operation. Then S is a Semigroup if the operation is associative.*

Definition 1.16 (Monoid). *Let S be a nonempty set with an operation. Then S is a Monoid group if the operation is associative and has an identity.*

Example: Consider the set Q as rational numbers and let $*$ be the operation on Q defined by $a * b = a + b - ab$.

a. Find 3 $*$ 4 and 2 $*$ (-5)

b. (i) Is (Q , $*$) a semigroup? (ii) Is it commutative?

c. Find the identity element of $*$

d. Show that (Q , $*$) is a monoid?

e. Which elements in Q, if any, have inverses and what are they?

Solution

a. Find 3 $*$ 4 and 2 $*$ (-5)

$$\begin{aligned} a * b &= a + b - ab \\ 3 * 4 &= 3 + 4 - (3 \times 4) \\ &= 3 + 4 - 12 \\ &= -5 \end{aligned}$$

$$\begin{aligned} a * b &= a + b - ab \\ 2 * (-5) &= 2 + (-5) - (2 \times (-5)) \\ &= 2 - 5 + 10 \\ &= 7 \end{aligned}$$

b. (i) Is (Q , $*$) a semigroup?

$$\begin{aligned} a * b &= a + b - ab \\ (a * b) * c &= (a + b - ab) * c \\ &= (a + b - ab) + c - (a + b - ab)c \end{aligned}$$

$$
\begin{aligned}
&= a + b - ab + c - ac - bc + abc
\end{aligned}
$$

$$
\begin{aligned}
a * (b * c) &= a * (a + b - ab) \\
&= a + (a + b - ab) - a(a + b - ab) \\
&= a + b + c - bc - ab - ac + abc \\
(a * b) * c &= a * (b * c)
\end{aligned}
$$

Hence (Q , $*$) is a semigroup.

(ii) Is it commutative?

$$
\begin{aligned}
a * b &= a + b - ab \\
&= b + a - ba \\
&= b * a
\end{aligned}
$$

Hence (Q , $*$) is a commutative semigroup.

c. Find the identity element of $*$

An element e is an identity element if $a * e = a$ for every $a \in Q$

$$
\begin{aligned}
a * e &= a \\
a + e - ae &= a \\
e - ae &= a - a \\
e(1 - a) &= 0 \\
e &= \frac{0}{1 - a} \\
e &= 0
\end{aligned}
$$

Therefore 0 is the identity element.

d. Show that (Q , $*$) is a Monoid?

By part b.(i) we have shown that the operation (Q , $*$) is associative hence leading into a Semigroup. Also by part (c.) we have also shown that the operation has an identity. Therefore by definition we conclude that the operation (Q , $*$) is a Monoid.

e. Which elements in Q, if any, have inverses and what are they?
An element x is an inverse of a if $a * x = 0$ where 0 is the identity element in part (c.)

$$\begin{aligned} a * x &= 0 \\ a + x - ax &= 0 \\ a &= ax - x \\ a &= x(a-1) \end{aligned}$$

$$x = \frac{a}{a-1}$$

If $a \neq 1$ then a has an inverse and it is $\dfrac{a}{a-1}$

1.2.2 Abelian , sub-group and Cyclic Group

Definition 1.17 (Group)**.** *Let G be a non-empty set with a binary operation. Then G is called a group if the following axioms holds:*

- *[G1] Associative law: For any a b c in G we have (ab)c=a(bc)*
- *[G2] Identity element: There exist an element e in G such that ae=ea=a for every a in G.*

- *[G3] Inverses: For each $a \in G$, $\exists$ an element $a^{-1} \in G$ (the inverse of a) such that $aa^{-1} = a^{-1}a = e$*

Definition 1.18 (Abelian group)**.** *A group G is said to be abelian (or commutative) if ab=ba for every a, b $\in$ G, that is if G satisfies commutative law.*

When G is abelian , the binary operation is denoted by + and G is said to be written additively.
The number of elements in a group G denoted by $\mid G \mid$ is called Order of G.

Definition 1.19 (Multiplicative inverse)**.** *The integers r and s are inverse if r $*$ s = 1.*

Definition 1.20 (Sub-group)**.** *Let H be a subset of a group G. Then H is called a subgroup of G if H itself is a group under operation of G*

Let G be any group and let a be any element of G. We can define $a^0 = e$ and $a^{n+1} = a^n \cdot a$. Let S denote the set of all the powers of a such that $S = \{\ldots, a^{-3},\ a^{-2},\ a^{-1},\ e\ , a\ , a^2, a^3 \ldots\}$. Then$S$ is a subgroup of G called the Cyclic group generated by a denoted by $gp(a)$.

The smallest positive integer m such that $a^m = e$ is called the <u>Order of a</u> denoted by $\mid a \mid$. If $\mid a \mid = m$ then the cyclic subgroup $gp(a)$ has m elements as follows:
$gp(a) = \{e,\ a,\ a^2,\ a^3,\ \ldots a^{m-1}\}$.

Definition 1.21 (Cyclic group)**.** *A group G is said to be Cyclic if it has an element a such that $G = gp(a)$.*

Example: Consider the group G= { 1, 2 ,3 ,4 ,5, 6} under multiplication modulo 7.

a. Find the multiplication table of G .

b. Find $2^{-1},\ 3^{-1}$ and 6^{-1}

c. Find the orders and subgroups generated by 2 and 3.

d. Is G cyclic?

Solution

a. Find the multiplication table of G .

To find $a{*}b$ in G, we find the remainder if when the product ab is divided by 7.

$*$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	**1**	3	5
3	3	6	2	5	**1**	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	**1**

b. Find $2^{-1},\ 3^{-1}$ and 6^{-1}

By definition integers r and s are inverse if r $*$ s = 1. Therefore

i. From multiplication table of G 2 $*$ 4 = 1. Hence $2^{-1} = 4$

ii. From multiplication table of G 3 $*$ 5 = 1. Hence $3^{-1} = 5$

iii.From multiplication table of G 6 $*$ 6 = 1. Hence $6^{-1} = 6$

c. Find the orders and subgroups generated by 2 and 3.

Recall by definition that $\mid a \mid$ is a smallest positive integer m such that $a^m = e$. Therefore we have: 2 $\Longrightarrow 2^1 = 2$, $2^2 = 4$, but $2^3 = 1$. Hence

i. $\mid 2 \mid$ =3

ii. $gp(2) = \{1, 2, 4\}$

3 $\Longrightarrow 3^1 = 3$, $3^2 = 2$, $3^3 = 6$, $3^4 = 4$ $3^5 = 5$ but $3^6 = 1$. Hence

i. $\mid 3 \mid$ =6

ii. $gp(3) = \{1, 2, 3, 4, 5, 6\}$

d. Is G cyclic?

For G to be Cyclic $G = gp(a)$. Since $G = \{1, 2, 3, 4, 5, 6\}$ and $gp(3) = \{1, 2, 3, 4, 5, 6\}$. We have $G = gp(3)$. Therefore G is Cyclic.

1.2.3 Permutation groups

Definition 1.22. *A one-to-one mapping σ of the set $\{$1 ,2 ,... n$\}$ onto itself is called Permutation (symmetric) group.*

Symmetric group S_n has n!= $n(n-1)\ldots 2\ .\ 1$. The symmetric group S_3 has 3!=6 elements as follows:

$$\begin{Bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{Bmatrix}, \begin{Bmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{Bmatrix}, \begin{Bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{Bmatrix}, \begin{Bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{Bmatrix}, \begin{Bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{Bmatrix}, \begin{Bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{Bmatrix}$$

1.2.4 Introduction to Rings and Fields

Definition 1.23. *Let R be a nonempty set with two binary operations, an operation of addition and an operation of multiplication. The R is called a ring if the following axioms are satisfied:*

- *[R1] For any $a,\ b,\ c \in R$ we have $(a+b)+c = a+(b+c)$*

- *[R2] $\exists$ an element $0 \in R$, called zero element , such that , for every $a \in R$, $a+0 = 0+a = a$.*

- *[R3] For each $a \in R$, $\exists$ an element $-a \in \mathbb{R}$, called negative of a, such that $a+(-a) = (-a)+a = 0$*

- *[R4] For any $a,\ b, \in R$ we have $a+b = b+a$*

- *[R5] For any $a,\ b,\ c \in R$ we have $(ab)c = a(bc)$*

- *[R6] For any $a,\ b,\ c \in R$ we have (i) $a(b+c) = ab+ac$ and $(ii) \quad (b+c)a = ba+ca$*

Definition 1.24. *R is called a commutative ring if $ab = ba$ for every $a, b \in \mathbb{R}$.*

R is called a ring with an identity element 1 if the element 1 has the property that $a.1 = 1.a = a$ for every element $a \in R$.
An element $a \in R$ is called a $unit$ if a has a multiplicative inverse thus $a^{-1} \in R$ such that $a.\ a^{-1} = a^{-1}.\ a$=1

Definition 1.25. *Field If A is a commutative ring with unity in which every non-zero element is invertible then A is called a Field.*

An element is invertible in a ring if there is some x in the ring such that $ax = xa = 1$

Definition 1.26. *(Polynomial over a field) Let K be a field. A polynomial f over K*

is an infinite sequence of elements from K in which all of them except a finite number are 0.

$$f(t) = a_n t^n + ...\ a_1 t\ +\ a_0$$

Example : Consider the ring $Z_1 0 = 0,\ 1,\ 2,\ ...9$ of the integers modulo 10.

a. Find the units of Z_{10}.

b. Find -3 , -8 , and 3^{-1}.

c. Let $f(x) = 2x^2 + 4x + 4$. Find the roots of $f(x)$ over Z_{10}.

Solution

a. Find the units of Z_{10}.

Recall that the unit of a ring R form a group under multiplication. And the units in Z_m are those integers which are relatively prime to m. Therefore those integers relatively prime to the modulus $m = 10$ are the units in Z_{10}. Hence we have 1, 3 , 7 and 9 as units in Z_{10}.

b. Find -3 , -8 , and 3^{-1}.

Recall that $-a$ in a ring R is the element such that $a + (-a) = (-a) + a = 0$.

i. Hence -3=7 since 3+7=7+3=0 in Z_{10}.

ii. Similary we have -8=2 since 2+8=8+2=0 in Z_{10}.

iii. Recall that a^{-1} in a ring is the element such that $a.\ a^{-1} = a^{-1}.\ a$=1. Hence $3^{-1} = 7$. Since 3.7=7.3=1 in Z_{10}.

c. Let $f(x) = 2x^2 + 4x + 4$. Find the roots of $f(x)$ over Z_{10}.

To find $f(x)$ we substitute each of the ten elements in Z_{10} into $f(x)$ to see which elements yields 0. We have $f(x) = 2x^2 + 4x + 4$.

$f(0) = 4$, $f(1) = 0$, $f(2) = 0$, $f(3) = 4$, $f(4) = 2$, $f(5) = 4$, $f(6) = 0$, $f(7) = 0$, $f(8) = 4$, $f(9) = 2$. The roots are 1, 2, 6 and 7

Example: Find the roots of $f(t)$ assuming that $f(t)$ has an integer root.$f(t) = t^3 + 2t^2 - 13t - 6$

Solution: Since the ideal coefficient is 1. Then the roots will be within ± 1,± 2, ± 3, ± 6. So $f(3)\ in\ t^3 + 2t^2 - 13t - 6$=0.Therefore $t = 3$ is a root and $t - 3$=0 if a factor. Dividing $t - 3$ in $f(t) = t^3 + 2t^2 - 13t - 6$ we get $(t - 3)(t^2 + 5t - 2)$. Using quadratic fprmular we have the following roots t=3 , $(\dfrac{-5 \pm \sqrt{17}}{2})$

Activity 1b

1. Consider the set N of positive integers and let $*$ denotes Least Common Multiple (L.C.M) operation on N
   (a) Find (i) $4 * 6$, (ii) $3 * 5$, (iii) $9 * 18$, (iv) $1 * 6$
   (b) Is $(N, *)$ a Semigroup? Is it Commutative?
   (c) Find the identity element of $*$
   (d) Which element in N if any have inverses and what are they?

2. Let G be a reduced residue system modulo 15 say G= { 1, 2 ,3 ,4 ,7, 8 , 11 , 13 , 14}. Then G is a group under multiplication modulo 15.
   (a) Find the multiplication table of G .
   (b) Find i. 2^{-1} ii. 7^{-1} and iii. 11^{-1}
   (c) Find the orders and subgroups generated by 2 , 7 and 11.
   (d) Is G cyclic?

3. Consider the ring $Z_{12} = \{0,\ 1,\ 2,\ ...11\}$ of the integers modulo 10.
   (a) Find the units of Z_{12}
   (b) Find the roots of $f(x) = x^2 + 4x + 4$ over Z_{12}
   (c) Find the associates of 2

4. Let $K = Z_8$. Find a possible roots of $f(t) = t^2 + 6t$

1.3 Summary

In this Unit, We have learnt about counting Techniques and its application toward Mathematics and Computer. We have talked about Product rule, Sum rule Pigeon-hole Principle and Exclusion-Inclusion principle as techniques for solving counting related problems. In the next unit, you will learn about Counting principles.

1.4 End of Unit Test

1. Find x and y given that $(2x,\ x+y) = (6,\ 2)$

2. Let $A = \{1, 2\}$ and $B = \{a, b, c\}$, solve for $A \times B$, A^2 and B^2

3. Given that vector $A = 3i + 2j - k$ and $B = 5i + 5j$. Find the Cross product?

4. Show that $a \times a = 0$ for any vector a in v_3

5. Calculate the cross product of two vectors with their scalar magnitude as $|a| = 2\sqrt{3}$ and $|b| = 4$ and the angle between them is $60°$

6. Find the roots of $f(t)$ assuming that $f(t)$ has an integer root $f(t) = t^3 + 2t^2 - 6t - 3$

7. Consider the ring $Z_{30} = \{0,\ 1,\ 2,\ ...11\}$ of the integers modulo 30.
   (b) Find -2, -7 and -11
   (c) Find 7^{-1},11^{-1} 26^{-1}

8. Given A= {1, 2, 3, 4} and B= $\{x, y, z\}$. Let R be the following relations from A to B: $R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$

(a) Determine the matrix of the relation.

(b) Draw the arrow diagram of R

(c) Find the inverse relation R^{-1} of R.

(d) Determine the domain and range of R.

8. Let A= {1, 2 ,3} , $B = \{a, b, c\}$ and $C = \{x, y, z\}$. Consider the following relations R and S from A to B and from B to C respectively.

$R = \{(1, b), (2, a), (2, c)\}$ and $S = \{(a, y), (b, x), (c, y), (c, z)\}$

(a) Find the composition relation $R \circ S$

(b) Find the matrices M_R , M_S and $M_{R \circ S}$ of the respective relations R, S and $R \circ S$.

(c) Compare $M_{R \circ S}$ to the product $M_R M_S$.

9. Consider the relation $R = \{(1, 1),\ (2\ , 2)\ , (2\ , 3), (3\ , 2), (4\ , 2), (4\ , 4)\}$ on $A = \{1,\ 2,\ 3, 4\}$. Draw its directed graphs.

10. Consider the set R as real numbers and let $*$ be the operation on R defined by $a * b\ \ = a + b + 2ab$.

(a) Find 2 $*$ 3 and 3 $*$ (-5)

(b) (i) Is (R , $*$) a semigroup? (ii) Is it commutative?

(c) Find the identity element of $*$

(d) Show that (Q , $*$) is a monoid?

(e) Which elements in R, if any, have inverses and what are they?

11. Consider G= { 1, 5,7 , 11 , 13 , 17}under multiplication modulo 18.

(a) Construct a the multiplication table of G .

(b) Find i. 5^{-1} ii. 7^{-1} and iii. 17^{-1}

(c) Find the orders and subgroups generated by 5 and 13.

(d) Is G cyclic?

Answers and Hints to Activities

Activity 1a

1. 64 relations from A to B

2. a = 2 and b = 5

3. (a) S and $A \times A$ are reflexive since (1, 1), (2,2), (3,3) exist in the relations

   (b) $S, \varnothing$ and $A \times A$ are symmetric

   (c) $R, S, \varnothing$ and $A \times A$ are transitive.

4. $A \times B$ = $\{(4\ ,7),\ (4\ ,8),\ (5,\ 7),\ (5\ ,8),(6\ ,7),(6,8)\}$

Activity 1b

1. (a)i. 12 (a)ii. 15 (a)iii. 18 and (a)iv. 6

   (b)i. Yes and (b)ii. Yes

   (c)1

   (d) has only one and it is its own inverse

2. (a)

$*$	1	2	4	7	8	11	13	14
1	1	2	4	7	8	11	13	14
2	2	4	8	14	**1**	7	11	13
4	4	8	1	13	2	14	7	11
7	7	14	13	4	11	2	1	8
8	8	1	2	11	4	13	14	7
11	11	7	14	2	13	**1**	8	4
13	13	11	7	1	14	**8**	4	2
14	14	13	11	8	7	**4**	2	1

(b) i. 8 (b)ii. 13 (b)iii.11

(c) i. $|\ 2\ |$ = 4 and gp(2)= $\{$ 1 ,2, 4, 8$\}$

(c)ii. $|\ 7\ |$ = 4 and gp(7)= $\{$ 1 ,4, 7, 13$\}$

(c)iii.$|\ 11\ |$=2 and gp(11)= $\{$ 1 ,11$\}$

(d) No, Since no elements generates G

3. (a) The units are $\{$ 1, 5, 7, 11$\}$

   (b) 4 and 10

   (c) $\{$2, 10$\}$

4. $f(t)$ has four roots. i.e 0 ,2 ,4 and 6.

1.5 Further Reading

1. Epp, S.S. (2003).*Discrete Mathematics with Applications*, 4th Edition. Boston: Richard Stratton.

2. Lipschutz, S., Lipson, M. (2005).*Discrete Mathematics*. Tata Mc Graw Hill.

3. Rosen, K.H., (2012). *Discrete Mathematics and its Applications*, New York: McGraw Hill.

unit 2

COUNTING TECHNIQUES

Introduction

In this unit, we will introduce the basic techniques for counting which will be used to address some counting problems arising throughout Mathematics and computer sciences.These methods serve as the foundation for almost all counting techniques. For example, We must count the successful outcomes of experiments and all possible outcomes of these experiments to determine probabilities of discrete events. We need to count the number of operations used by an algorithm to study its time complexity.

Intended Learning Outcomes

By the end of this Unit, you should be able to:

a. Define Counting techniques.

b. Demonstrate understanding and ability to solve problems using product rule technique.

c. Compute and apply the sum rule in counting related problems.

d. Demonstrate understanding on the application of the Pigeonhole principle and Exclusion-Inclusion principle.

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed

below:
Counting functions, Product rule, Sum rule, Exclusion-Inclusion Principle Generalised Exclusion-Inclusion Principle , Pigeonhole Principle and Generalised Pigeonhole Principle.

BASIC COUNTING PRINCIPLES

Reflection : Think about any counting principle that can be applied to solve problems in Math and Computer.

We will present the following basic counting principles : product rule, the sum rule, Inclusion and Exclusion principle and the Pigeonhole principle . Then we will show how they can be used to solve many different counting problems.

2.1 THE PRODUCT RULE

This is a procedure that can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and n_2 ways to do the second task then there are $n_1.n_2$ ways to do the procedure.
Examples 1-3 show how product rule is used.

Example 1 Suppose a College has 3 different history courses, 4 different literature courses and 2 different sociology courses.How many ways can a student choose one of each kind of courses?

Solution: The number m of ways a student can choose one of each kind of courses is:

$$\begin{aligned} m &= n_1 \ . \ n_2 \ .n_3 \\ &= 3 \ . \ 4. \ 2 \\ &= 12 \end{aligned}$$

Example 2: A new company with just two employees Alice and Bob rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

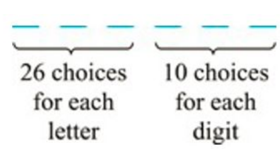
Solution: The procedure for assigning offices to these two employees consists of assigning an office to Bob , which can be done in 12 ways, then assigning an office to Alice different from the office assigned to Bob, which can be done in 11 ways. By product rule there are 12 . 11 = 132 ways to assign offices to these two employees.

Example 3: There are 32 microcomputers in a computer center. Each microcomputer has 24 ports. How many different ports to a microcomputer in the center are there?

Solution: The procedure for choosing a port consists of two tasks, first picking a microcomputer then picking a port of this microcomputer. Because they are 32 ways to choose the microcomputer and 24 ways to choose the port no matter which microcomputer has been selected, therefore the product rule shows that there are 32 . 24 = 768 ports.

Example 4: How many bit strings of length seven are there?
Solution: Each of the seven bits can be chosen in two ways , because each bit is either 0 or 1. Therefore the the product rule shows there are a total of 2^7 = 128 different bit strings of length seven.

Example 5: How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits?
Solution: There are 26 choices for each of the three letters and ten choices for each of the three digits.Hence by product rule there are a total of 26 . 26 . 26 . 10 . 10 . 10 = 17,576,000 different possible license plates.



2.1.1 Counting Functions

Example 6: How many functions are there from a set with m elements to a set with n elements?

Solution: Since a function represents a choice of one of the n elements of the codomain for each of the m elements in the domain, hence the product rule tells us that there are $n\ .\ n.\ \ ..\ .\ n = n^m$ such functions from a set with m elements to one with n elements. $i.e$ there are 7^4 different functions from a set with four elements to a set with seven elements.

2.1.2 Counting One-to-One Functions

Example7 How many one-to-one functions are there from a set with m elements to one with n elements?

Solution: Suppose the elements in the domain are a_1, a_2,. . . , a_m. There are n ways to choose the value of a_1 and $n-1$ ways to choose a_2, etc. The product rule tells us that there are $n(n-1)\ (n-2)...(n-m+1)$ such functions.

2.1.3 Product Rule in Terms of Sets

If $A_1, A_2, ..., A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements of each set. This means that the task of choosing an element in the Cartesian product $A_1 \times A_2 \times ... \times A_m$ is done by choosing an element in A_1, an element in A_2 ,..., and an element in A_m. Therefore by the product rule, it follows that:

$$\begin{aligned} n(A_1 \times A_2 \times ... \times A_m) &\iff \mid A_1 \times A_2 \times ... \times A_m \mid \\ &= \mid A_1 \mid . \mid A_2 \mid . \mid A_3 \mid ... \mid A_m \mid \end{aligned}$$

Activity 2a

1. Suppose a book shelf has 5 Algebra texts , 3 Discrete texts, 6 Calculus texts and 4 Cryptography texts. How many ways can a student choose one of each type of the text.

2. How many one-to-one functions are there from a set with 4 elements to one with 7 elements?

2.2 THE SUM RULE

Suppose some task A can be done in m ways and a second task B can be done in n ways, and suppose that both tasks cannot be done simultaneously, then by sum rule there are $m + n$ ways of doing a task A or B.
The following examples illustrate how sum rule is used.
Example: Suppose a College has 3 different history courses, 4 different literature courses and 2 different sociology courses.How many ways can a student choose just one of the courses?

Solution: The number n of ways a student can choose just one of the courses is:

$$\begin{aligned} n &= m_1 + m_2 + m_3 \\ &= 3 + 4 + 2 \\ &= 9 \end{aligned}$$

Example: The mathematics department must choose either a student or a faculty member as a representative for a university committee. How many choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student.

Solution: There are 37 ways to choose a member of Mathematics faculty and there are 83 ways to choose a student who is a Mathematics major. Choosing a member of Mathematics faculty is never the same as choosing a student who is a Mathematics major because no one is both a faculty member and a student. By the sum rule it follows that there are 37 + 83 = 120 possible ways to pick a representative.

Example: A student can choose a computer project from one of the three lists. The lists contains 23 ,15 and 19 possible projects , respectively. No project is more than one list. How many possible projects are there to choose from?

Solution: The student can choose a project by selecting a project from the first list, second list and third list. Because no project is on more than one list, by sum rule there are 23+15+19=57 ways to choose a project.

2.2.1 The Sum Rule in terms of Sets

The sum rule can also be phrased in terms of sets. If $A_1, A_2, ..., A_m$ are disjoint finite sets, then the number of elements in the union of these sets is the sum of the numbers of elements in the sets. This implies that there are $\mid A_i \mid$ ways to choose an element from $\mid A_i \mid$ for $i = 1, 2, ...m.$ By the sum rule , the number of ways to choose an element from one of the sets, which is the number of the elements in the union is given as follows:

$$| A_1 \cup A_2 \cup ... \cup A_m | \quad = \quad | A_1 | + | A_2 | + | A_3 | ...+ | A_m |$$

2.2.2 Combining the Product and the Sum Rule

Combining the sum and product rule allows us to solve more complex problems.

Example: Suppose statement labels in a programming language can be either a single letter or a letter followed by a digit. Find the number of possible labels.

Solution:
Single upper case letter = 26
Upper case letter **followed** by a digit= 26.10
Single upper case letter **or** Upper case letter **followed** by a digit implies combination of product and sum rule. Hence there are 26 + 26.10= 286 possible labels.

Counting Password

Example: Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

solution Let P be the total number of passwords, and let P_6, P_7, and P_8 be the passwords of length 6, 7, and 8.
By the sum rule P = P_6+ P_7 + P_8. To find each of P_6, P_7, and P_8 , we find the number of passwords of the specified length composed of letters and digits and subtract

the number composed only of letters. We find that:

$$\begin{aligned} P_6 &= (letters + digits)^6 - (letters)^6 \\ &= 36^6 - 26^6 \\ &= 2,176,782,336 - 308,915,776 \\ &= 1,867,866,560. \end{aligned}$$

Similarly , it can also be shown that

$$\begin{aligned} P_7 &= (letters + digits)^7 - (letters)^7 \\ &= 36^7 - 26^7 \\ &= 78,364,164,096 - 8,031,810,176 \\ &= 70,332,353,920. \end{aligned}$$

and

$$\begin{aligned} P_8 &= (letters + digits)^8 - (letters)^8 \\ &= 36^8 - 26^8 \\ &= 2,821,109,907,456 - 208,827,064,576 \\ &= 2,612,282,842,880. \end{aligned}$$

Therefore, P = P_6+ P_7 + P_8 = 2,684,483,063,360.

Activity 2b

1. Suppose a book shelf has 5 Algebra texts , 3 Discrete texts, 6 Calculus texts

and 4 Cryptography texts. How many ways can a student choose one of the text.

2. Suppose statement labels in a programming language can be either a sequence of two uppercase letters or an uppercase letter followed by three digits. Find the number of possible labels.

2.3 THE INCLUSION-EXCLUSION PRINCIPLE

If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the different ways.

Theorem 1. *Suppose A and B are finite sets. Then $A \cup B$ and $A \cap B$ are finite and*

$$n(A \cup B) \quad = \quad n(A) + n(B) - n(A \cap B).$$

That is , we find the number of elements in A or B (or both) by first adding $n(A)$ and $n(B)$ (inclusion) and then subtracting $n(A \cap B)$ (exclusion) since its elements were counted twice.

Example: Suppose a list A contains the 30 students in a Mathematics class, and a list B contains the 35 students in an English class, and suppose there are 20 names on both lists. Find the number of students on list A or B (or both).

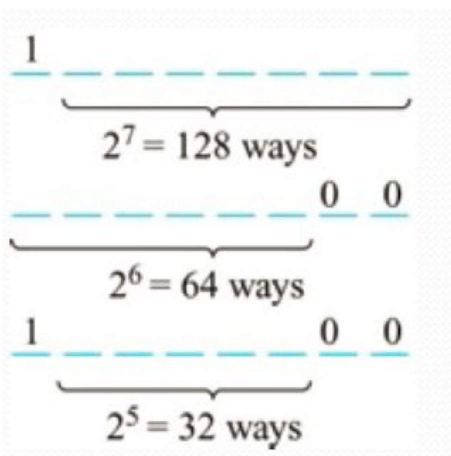
Solution: Since we seek to find $n(A \cup B)$. This means we are to combine the two lists and then cross out the 20 names which appear twice. By Inclusion and Exclusion Principle we have

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B). \\ &= 30 + 35 - 20 \\ &= 45 \end{aligned}$$

Example : (**Counting bit of strings**)

How many bit of strings of length eight either start with a 1 bit or end with bits 00?

Solution



• Number of bit of strings of length eight that start with 1 bit = $2^7 = 128$ ways.

• Number of bit of strings of length eight that ends with bit 00 = $2^6 = 64$ ways.

• Number of bit of strings of length eight that start with 1 bit and end with bits 00 = $2^5 = 32$ ways.

• Hence by exclusion-inclusion principle:

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B). \\ &= 128 + 64 - 32 \\ &= 160 \text{ ways} \end{aligned}$$

We can also apply the result in theorem 1 to obtain similar formula for three sets.

Theorem 2. *For any finite sets A, B, C we have*

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$

The theorem 2 states that we **include** $n(A)$, $n(B)$, $n(C)$, we **exclude** $n(A \cap B)$, $n(A \cap C)$, $n(B \cap C)$ and finally **include** $n(A \cap B \cap C)$.

Example: Find the number of Mathematics students at College taking at least one of the Languages, French ,German and Russians, given the following data:

65 study French	20 study French and German
45 study German	25 study French and Russian
42 study Russian	15 study German and Russian
	8 study all three languages

Solution: We want to find $n(F \cup G \cup R)$ where F, G and R denotes the sets of students studying French , German and Russian, respectively. By Inclusion-exclusion Principle,

$$\begin{aligned} n(F \cup G \cup R) &= n(F) + n(G) + n(R) - n(F \cap G) - n(F \cap R) - n(G \cap R) + n(F \cap G \cap R). \\ &= 65 + 45 + 42 - 20 - 25 - 15 + 8 \\ &= 100 \end{aligned}$$

There are 100 possible students who study at least one of the three language.

2.3.1 The Generalised Inclusion - Exclusive Principles

Now suppose we have any finite number of finite sets, say ,A_1, A_2, ... A_m. Let s_k be the sum of the cardinalities $n(A_{i1}$, $\cap$ A_{i2} $\cap$... $\cap A_{ik}$) of all possible k-tuple intersections of the given m sets. By a general Inclusion-Exclusion Principle we have the following :

$$n(A_1 \cup A_2 \cup ... \cup A_m) \quad = \quad s_1 - s_2 + s_3 - ... + (-1)^{m-1} s_m$$

Activity 2c

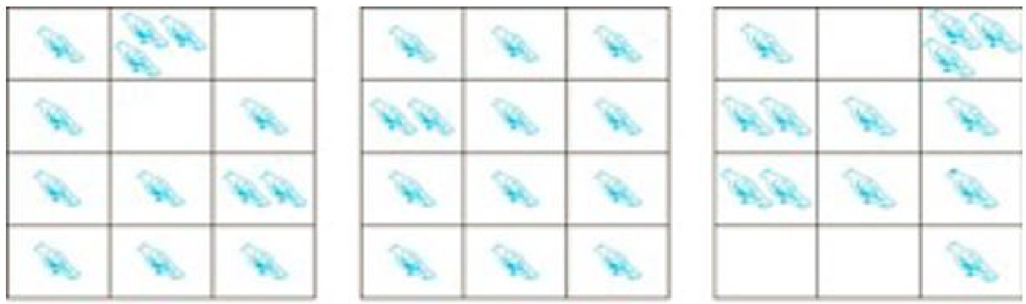
1. There are 22 Female students and 18 Male students in a classroom. Find the total number t of student.

2. Suppose among 32 people who save paper or bottles (or both) for recycling , there are 30 who save paper and 14 who save bottles. Find the number of m people who : (i) Save both , (ii) save only paper and (iii) only bottles.

3. A Computer company receives 350 applicants from computer graduates for a job planning a line of new job servers. Suppose that 220 of these people majored in computer science, 147 in business and 51 majored both in computer science and in business. How many of these applicants Majored neither in computer science nor in business?

2.4 THE PIGEONHOLE PRINCIPLE

Suppose that a flock of 20 pigeons flies into a set of 19 pigeonholes to roost, then it means that one of the pigeonholes must have more than 1 pigeon. Now lets look on the following diagram which illustrates the pigeon hole rule.



(a) (b) (c)

Theorem 3 (The Pigeonhole Principle). *It states that If k is a positive integer and k + 1 objects (pigeons) are placed into k boxes (pigeonholes), then at least one box (pigeonhole) contains two or more objects (pigeons).*

Proof: We use a proof by contraposition. Suppose none of the k boxes has more than one object. Then the total number of objects would be at most k. This contradicts the statement that we have k + 1 objects.

This principle can be applied to many problems where we want to show that a given situation can occur.Examples 1-3 shows how pigeonhole principle is used.

Example1: Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

Example 2 : In any group of 27 English words, there must be at least two that begin with the same letter , because there are 26 letters.

Example 3 : How many students must be in a class to guarantee that at least two students receive the same score on the final exam,if the exams is graded on a scale from 0 to 100 points.

Solution: There are 101 possible scores on the final exams. The pigeonhole principle shows that among any 101 students there are at least 2 students with the same score.

2.4.1 The Generalized Pigeonhole Principle

We now present the theorem which defines the generalised pigeonhole principle. The proof to this theorem is left as an exercise

Theorem 4. *The Generalized Pigeonhole Principle states that If N objects are placed into k boxes, then there is at least one box containing at least* $\lceil N/k \rceil$ *objects.*

Example 4 Among 100 people , how many were born in the same month?

Solution: Since $N = 100$ and $K = 12$ then the Generalized Pigeonhole Principle tells us that there are at least $\lceil N/k \rceil$ = $\lceil 100/12 \rceil$ = 9 who were born in the same month.

A common type of Generalized Pigeonhole Principle problems asks for minimum number of objects such that at least r of these objects must be in one of K boxes when these objects are distributed among the boxes. When we have N objects, the **GPhP** tells us that there must be at least r objects in one of the boxes.Therefore the minimum number of objects is the smallest integer $N = K.(r-1)+1$ satisfying the inequality $\lceil N/k \rceil \geq r$.

Example 5 :

Find the minimum number of students in a class to be sure that at least three of them were born in the same month.

Solution: Since we have $K = 12$ months and $r = 3$, therefore the **GPhP** tells us that the minimum number of students is the smallest integer $N = K(r-1)+1$. Hence among any 12.(3-1) + 1 = 25 students, three of them were born in the same month.

Example 6 : Find the minimum number of elements that one needs to take from the set $S =${1 , 2 , 3 , 4, ... 9 } to be sure that two of the numbers add up to 10.

Solution: Here the pigeonholes are the five sets {1,9} {2,8} {3,7} {4,6} {5}. Since K= 5 and r=2, then $N = K(r-1)+1$ implies 5.(2-1) +1=6. Therefore any choice of six element(pigeons) of S will guarantee that two of the numbers add up to 10.

Activity 2d

1. Find the minimum number of students needed to guarantee that five of them belong to the class (Freshman, Sophomore , Junior , Senior) .

2. Find the minimum number n of integers to be selected from S= $\{1, 2, 3, 4, ...9\}$ so that the sum of two n integers is even.

2.5 Summary

In this Unit, We have learnt about counting Techniques and its application toward Mathematics and Computer. We have talked about Product rule, Sum rule Pigeon-hole Principle and Exclusion-Inclusion principle as techniques for solving counting related problems. In the next unit, you will learn about recursion and recurrences .

2.6 End of Unit Test

1. Out of 196 students who registered for a Discrete Mathematics course at MUST , How many students were born in the same month?

2. What is the minimum number of students, each of whom comes from one of the 28 districts in Malawi, who must be enrolled in a university to guarantee that there are at least 12 who come from the same district?

3. An office building contains 29 floors and has 56 offices on each floor. How many offices are in the building?

4. A student can choose a computer project from one of the three lists. The three lists contains 23, 25 and 19 possible projects respectively. No project is more than one list. How many possible projects are there to choose from?

5. A particular brand of shirt comes in 12 colours, has a male version and a female version, and comes in three sizes (small, medium, large) for each sex. How many different types of this shirt are made?

6. How many bits of strings of length either begin with two 0s or end with three 1s?

7. How many license plates can be made using either two letters followed by four digits or two digits followed by four letters.

8. How many different functions are there from a set with 10 elements to sets with 3 elements.

9. How many numbers must be selected from the set $\{ 1 ,2 ,3, 4 ,5, 6\}$ to guarantee that at least one pair of these numbers add up to 7.

10. How many one-to-one functions are there from a set with five elements to set with six elements.

Answers and Hints to Activities

Activity 1a

1. 360 ways

2. 840

Activity 1b

1. 18 ways

2. 26676

Activity 1c

1. 40

2.

   i. 12

   ii. 18

   iii. 2

3. 34 Applicants

2.7 Further Reading

1. Epp, S.S. (2003). *Discrete Mathematics with Applications*, 4th Edition. Boston: Richard Stratton.

2. Lipschutz, S., Lipson, M. (2005). *Discrete Mathematics*. Tata Mc Graw Hill.

3. Rosen, K.H., (2012). *Discrete Mathematics and its Applications*, New York: McGraw Hill.

unit 3

RECURSION AND RECURRENCES

Introduction

In this unit, we will study a variety of counting problems that can be modeled using recurrence relations. These problems are solved by finding the relationships between the terms of sequence. We will also develop the methods for finding explicit formula for terms of sequences that satisfy certain types of recurrence relations. We will also investigate the solutions of homogeneous linear recurrence relations with constant coefficients.

Intended Learning Outcomes

By the end of this Unit, you should be able to:

a. Define Recurrence relations.

b. Demonstrate understanding and ability to solve problems involving recurrence relations

c. Demonstrate understanding to solve homogeneous linear recurrence relations problems .

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:
Recurrence relation, Homogeneous, Initial condition, closed formula .

Reflection : Think about other counting principle that can be applied to solve problems in Math and Computer.

We will present another counting principle which is the recursive and recurrence relations. Then we will show how we can be use it to solve some counting problems that were not possible to be solved using other counting techniques presented in previous chapter.

3.1 RECURRENCE RELATIONS

Definition 3.1. *A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n interms of one or more of the previous terms $a_0,\ \ a_1, . . .a_{n-1}$ for integers n with $n \geq n_0$ where n_0 is a nonnegative integer.*

Asequence is called a *solution* of recurrence relation if its terms satisfy the recurrence relation. The *Initial condition* is required to precede the first terms where relations take effect.

Example 3.1: Let a_n be the sequence that satisfies the recurrence relation $a_n =$

$a_{n-1} + 6$ for $n \geq 1$ given that $a_0 = 3$. Find a_1 ,a_2, a_3 ... a_{100}

Solution:

$$\begin{aligned} a_0 &= 3 \\ a_1 &= a_0 + 6 = 3 + 6 = 9 \\ a_2 &= a_1 + 6 = 9 + 6 = 15 \\ a_3 &= a_2 + 6 = 15 + 6 = 21 \end{aligned}$$

We can an equation a+nd. Since

$$\begin{aligned} a_4 &= 3 + 4.6 = 27 \text{ . Therefore} \\ a_{100} &= a + n.d = 3 + (100 \times 6) \\ &= 603 \end{aligned}$$

3.1.1 Solving recurrence relation

When solving recurrence relation we aim at finding a non-recursive formula to calculate a_n where the solution is called a closed formula. We can also use Iteration method involving substitution and back tracking methods to solve these recurrence relation. In the following example we are going to apply these methods.

Example 3: Let a_n be the sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n \geq 1$ with initial condition $a_0 = 2$. Use Iteration method involving substitution or back tracking method to solve for a_1 ,a_2, a_3 , a_4 ... a_n

Solution: By Iteration method involving substitution we have

$$
\begin{aligned}
a_0 &= 2 \\
a_1 &= 2+3 \\
a_2 &= a_1+3 = (2+3)+3 = 2(3+3) = 2+2(3) \\
a_3 &= a_2+3 = 2+2(3)+3 = 2+3(3) \\
a_4 &= 2+4(3) \\
&\vdots \\
a_n &= 2+n(3) \\
a_n &= 2+3n
\end{aligned}
$$

The use of back tracking is left for you as an exercise.

Activity 2a

1. Find the next two elements of the following recurrence relations with their initial conditions.

   (a) $a_n = 2a_{n-1}an - 2 + n^2$; $a_1 = 1$, $a_2 = 2$

   (b)$a_n = 2a_{n-1} + 5an - 2 - 6n_{n-3}$; $a_1 = 1$, $a_2 = 2$ and a_3=1

   (c)$a_n = na_{n-1} + 3an - 2$; $a_1 = 1$, $a_2 = 2$

3.2 LINEAR RECURRENCES

In linear recurrence , each term of a sequence is a Linear function of earlier terms in the sequence. Now we will discuss in details the two types of Linear recurrences. We therefore start by giving the definition of Linear homogeneous recurrence.

Definition 3.2. *A linear homogeneous recurrence relation of degree K with constant* *coefficients is a recurrence relation of the form* $a_n = C_1a_{n-1} + C_2a_{n-2} + \cdots\ C_ka_{n-k}$ *where* C_1 *,* C_2 *,* $\cdots C_k$ *are real numbers and* $C_k \neq$ *0.*

a_n is expressed in terms of the previous K terms of the sequence and therefore its degree is K. This recurrence include k initial conditions such that $a_0 = C_0$, $a_1 = C_1$ $\cdots\ a_k = C_k$.

3.2.1 Solving second order homogeneous Linear recurrence relations

Consider a homogeneous second-order recurrence relation with constant coefficient which has the form $a_n = sa_{n-1} + ta_{n-2}$ or $a_n - sa_{n-1} - ta_n - 2 = 0$ where s and t are constants with $t \neq 0$. We associate the following quadratic polynomial with the above recurrence relation $\triangle(x) = x^2 - sx - t$. This polynomial $\triangle(x)$ is called $characteristics\ polynomial$ of the above recurrence relation and the roots are called $Characteristics\ roots$.

Theorem 5. *Suppose the characteristic polynomial* $\triangle(x) = x^2-sx-t$ *of the recurrence relation* $a_n = sa_{n-1} + ta_{n-2}$ *has distinct roots* r_1 *and* r_2*. Then the general solution of the recurrence relation is denoted by* $a_n = C_1r_1^n + C_2r_2^n$ *where* C_1 *and* C_2 *are arbitrary constants.*

Example Consider the homogeneous recurrence relation $a_n = 2a_{n-1} + 3a_{n-2}$ with initial condition a_0=1 , $a_1 = 2$.

a. Find the general solution

b. Find the unique solution with the given conditions.

Solution

a. Find the general solution

First we find the characteristic polynomial $\triangle(x)$ and its roots r_1 and r_2.

$$\begin{aligned} \triangle(x) &= x^2 - 2x - 3 = 0 \\ &= (x-3)(x+1) = 0 \end{aligned}$$

The roots r_1=3 and r_2=-1. Since the roots are distinct then by theorem 1 above the general solution is $a_n = C_1 3^n + C_2(-1)^n$

b. Find the unique solution with the given conditions.

Unique solution is obtained by finding C_1 and C_2 using initial conditions. For n=0 and $a_0 = 1$ we get a_0= $C_1 3^0 + C_2(-1)^0 = 1 \Rightarrow C_1 + C_2 = 1$

For n=1 and $a_2 = 1$ we get a_1= $C_1 3^1 + C_2(-1)^1 = 2 \Rightarrow 3C_1 - C_2 = 2$

Solving the system of two equations in the unknown C_1 and C_2 yields $C_1 = \frac{3}{4}$ and $C_2 = \frac{1}{4}$. The unique solution of the given recurrence relation with the given initial conditions $a_0 = 1$ and $a_1 = 2$ is

$$\begin{aligned} a_n &= \frac{3}{4}3^n + \frac{1}{4}(-1)^n \\ &= \frac{3^{n+1} + (-)^n}{4} \end{aligned}$$

Theorem 6. *Suppose the characteristic polynomial $\triangle(x) = x^2 - sx - t$ of the recurrence relation $a_n = sa_{n-1} + ta_{n-2}$ has only one root r_0 (no distinct roots). Then the general solution of the recurrence relation where C_1 and C_2 are arbitrary constants is denoted by $a_n = C_1 r_0^n + C_2 n r_0^n$.*

The constants C_1 and C_2 may be uniquely computed using initial conditions.

Example Consider the following homogeneous recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial condition a_0=3 , $a_1 = 27$.

a. Find the general solution

b. Find the unique solution with the given conditions.

Solution

a. Find the general solution

   First we find the characteristic polynomial $\triangle(x)$ and its roots r_1 and r_2.

$$\begin{aligned}\triangle(x) &= x^2 - 6x + 9 = 0 \\ &= (x-3)(x-3) = 0 \\ &= (x-3)^2\end{aligned}$$

   Thus $\triangle(x)$ has only one root r_0=3. Therefore by theorem 2 the general solution of the recurrence relation is a_n= $C_1 3^n + C_2 n 3^n$

b. Find the unique solution with the given conditions.

   The unique solution is obtained by finding C_1 and C_2 using the initial conditions. For $n = 1$ and $a_1 = 3$ we get $C_1 3^1 + C_2(1)(3)^1 = 3$ or $3C_1 + 3C_2 = 3$. For $n = 2$ and $a_2 = 27$ we get $C_1 3^2 + C_2(2)(3)^2 = 27$ or $9C_1 + 18C_2 = 27$. Solving the system of the two equations in the unknowns C_1 and C_2 yields: $C_1 = -1$ and $C_2 = 2$. The unique solution of the given recurrence relation with the given initial conditions

$$a_n = -3^n + 2n3^n$$

$$= \quad 3^n(2n-1)$$

3.2.2 Solving General Homogeneous Linear Recurrence Relations

Consider now a general linear homogeneous kth-order recurrence relation with constant coefficient which has the form $a_n = C_1a_{n-1} + C_2a_{n-2} + C_3a_{n-3} \cdots + C_ka_{n-k} = \sum_{i=1}^{k} C_ia_{n-1}$ where $C_1, C_2, \cdots C_k$ are constants with $C_k \neq 0$. The characteristic polynomial $\triangle(x)$ of the recurrence relation

$$\begin{aligned}
\triangle(x) &= x^k - C_1x^{k-1} - C_2x^{k-2} - C_3x^{k-3} - \cdots - C_k \\
&= x^k - \sum_{i=1}^{k} C_ix_{k-1}
\end{aligned}$$

Example Consider the following third order homogeneous recurrence relation $a_n = 11a_{n-1} - 39a_{n-2} + 45a_{n-3}$ with initial conditions a_0=5 , $a_1 = 11$ and $a_2 = 25$.

a. Find the general solution

b. Find the unique solution with the given conditions.

Solution

a. Find the general solution

   First we find the characteristic polynomial

$$\begin{aligned}
\triangle(x) &= x^3 - 11x^2 + 39x - 45 \\
&= (x-3)^2(x-5)
\end{aligned}$$

   $\triangle(x)$ has two roots $r_1 = 3$ of multiplicity of 2 and $r_2 = 5$ of multiplicity of 1.

The general solution of the recurrence relation is denoted by

$$\begin{aligned} a_n &= C_1(3^n) + C_2 n(3^n) + C_3(5^n) \\ &= (C_1 + C_2 n)(3^n) + C_3(5^n) \end{aligned}$$

b. Find the unique solution with the given conditions.

The unique solution is obtained by finding C_1, C_2 and C_3 using the initial conditions. For $n = 0$ and $a_0 = 5$ we get $C_1 + C_3 = 5$.
For $n = 1$ and $a_1 = 11$ we get $3C_1 + 3C_2 + 5C_3 = 11$.
For $n = 2$ and $a_2 = 25$ we get $9C_1 + 18C_2 + 25C_3 = 25$.
Solving the three equations for unknown $C_1 = 4 \quad C_2 = -2 \quad C_3 = 1$. Therefore the unique solution of recurrence relation with given initial conditions
$a_n = (4 - 2n)(3^n) + 5^n$.

3.2.3 Non-homogeneous linear recurrence relations

A linear non-homogeneous recurrence relation with constant coefficient is a recurrence relation of the form $a_n = C_1 a_{n-1} + C_2 a_{n-2} + C_3 a_{n-3} \cdots + C_k a_{n-k} + F(n)$ where $C_1, C_2, \cdots C_k$ are real numbers and $F(n)$ is a function of n no identically zero.

The recurrence relation $a_n = C_1 a_{n-1} + C_2 a_{n-2} + C_3 a_{n-3} \cdots + C_k a_{n-k}$ is called $associated\ homogeneous\ recurrence\ relation$. This recurrence includes k initial conditions. $a_0 = C_0 \quad a_1 = C_1 \cdots a_k = C_k$
Some of the examples of non-homogeneous linear recurrence relation include

$$\begin{aligned} i.\ a_n &= a_{n-1} + 2^n \\ ii.\ a_n &= a_{n-1} + an - 2 + n^2 + n + 1 \\ iii.\ a_n &= a_{n-1} + an - 2 + a_{n-3} + n! \end{aligned}$$

In non-homogeneous Linear recurrence relation with constant coefficient, every solution is the sum of the $particular\ solution$ and the $associated\ linear\ homogeneous\ recurrence\ relations$ i.e $\quad a_n^{(h)} + a_n^{(p)}$

Theorem 7. *If $\{a_n^{(p)}\}$ is a particular solution of non-homogeneous linear recurrence relation with constant coefficients $a_n = C_1a_{n-1} + C_2a_{n-2} + C_3a_{n-3} \cdots + C_ka_{n-k} + F(n)$ then every solution is of the form $a_n^{(h)} + a_n^{(p)}$ where $a_n^{(h)}$ is a solution of the associated recurrence relation $a_n = C_1a_{n-1} + C_2a_{n-2} + C_3a_{n-3} \cdots + C_ka_{n-k}$.*

Example: Find all solutions of the recurrence relation $y_n = 3y_{n-1} + 2n$ with initial condition $y_1 = 3$

Solution

We start by finding the associated homogeneous linear equation

$$\begin{aligned} y_n &= y_n^{(h)} + y_n^{(p)} \\ y_n^{(h)} &= 3y_{n-1} \quad \text{since r-3=0 then r=3 (one root)} \\ y_n^{(h)} &= C.3^n \end{aligned}$$

The particular solution

Since $F(n) = 2n$ is a polynomial of degree n. A guess form (reasonable trial solution) is a linear function in n, say $p_n = Cn + d$, where c and d are constants. Then $y_n = 3y_{n-1} + 2n$ becomes

$$Cn + d = 3[C(n-1) + d] + 2n$$

$$
\begin{aligned}
&= 3(Cn - c + d) + 2n \\
Cn + d &= 3Cn - 3c + 3d + 2n \\
0 &= 3Cn - 3c + 3d + 2n - cn - d \\
0 &= (2Cn + 2)(-3c + 2d)
\end{aligned}
$$

It follows that Cn+d is a solution if and only if 2c+2=0 or 2d-3c=0. Hence c=-1 or d= -$\frac{3}{2}$. So the Particular solution is

$$
\begin{aligned}
P_n &= Cn + d \\
y_n^{(P)} &= -n - \frac{3}{2} \\
y_n &= y_n^{(h)} + y_n^{(p)} \\
&= C.3^n + (-n - \frac{3}{2})
\end{aligned}
$$

We solve the constant term using the initial condition $y_1 = 3$ when n=1

$$
\begin{aligned}
Y_1 &= C.3^1 + (-n - \frac{3}{2}) = 3 \\
3C &= 3 + 1 + \frac{3}{2} \\
C &= \frac{11}{6}
\end{aligned}
$$

Therefore the solution is $y_n = -n - \frac{3}{2} + (\dfrac{11}{6})3^n$

Activity 2b

1. Find all solutions for the recurrence $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$

2. Find all solutions for the recurrence $a_{n+2}3a_{n+1} + 2a_n = 3^n$

3. Find the solution to recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with initial conditions $a_0 = 1$, a_1=-2 and $a - 2 = -1$

3.3 Summary

In this Unit, We have learnt about counting Techniques and its application toward Mathematics and Computer. We have talked about recurrence relation, homogeneous and non homogeneous linear recurrence relation. In the next unit, you will learn about Generating functions .

3.4 End of Unit Test

1. Show that the sequence $a_n = (-4)^n$ is a solution to the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$

2. Consider the second order homogeneous recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ with initial conditions $a_0 = 2$ and $a_1 = 7$
   (a) Find the next three terms of the sequence.
   (b)Find the general solution
   (c) Find the unique solution with the given initial conditions.

3. Consider the third order homogeneous recurrence relation $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$ with initial conditions $a_0 = 3$ $a_1 = 4$ and $a_2 = 12$
   (a)Find the general solution
   (b) Find the unique solution with the given initial conditions.

4. Determine whether the following recurrence relations are linear homogeneous recurrence relation with constant coefficient.
   (a) $P_n = (11.1)P_{n-1}$
   (b) $a_n = a_{n-1} + a_{n-2}^2$

(c)$F_n = f_{n-1} + f_{n-2}$

(d) $a_n = a_{n-6}$

Answers and Hints to Activities
Activity 3a

1. (a) a_3=13 and a_4= 68

   (b) a_4=4 and a_5= -37

   (c) a_3=9 and a_4= 42

Activity 3b

1. $C_1.2^n + c_2.3^n + (\frac{49}{20}).7^n$

2. $C_1(-2^n) + c_2(-1)^n + (\frac{1}{20}).3^n$

3. $a_n = (1 + 3n - 2n^2)(-1)^n$

3.5 Further Reading

1. Epp, S.S. (2003).*Discrete Mathematics with Applications*, 4th Edition. Boston: Richard Stratton.

2. Lipschutz, S., Lipson, M. (2005).*Discrete Mathematics*. Tata Mc Graw Hill.

3. Rosen, K.H., (2012). *Discrete Mathematics and its Applications*, New York: McGraw Hill.

unit 4

GENERATING FUNCTIONS

Introduction

Generating functions offer a powerful framework for studying and solving problems in discrete Mathematics. They provide a concise representation of sequences and allow the application of algebraic techniques to derive meaningful results and insights. Therefore In this unit, we will discuss some basic concepts of generating functions and look into application of generating functions in solving recurrence relation problems

Intended Learning Outcomes

By the end of this Unit, you should be able to:

a. distinguish generating sequence and generating functions .

b. build generating functions

c. solve generating functions using multiply shift and subtract technique

d. solve recurrence relations with generating functions

 **Key Terms**

Ensure that you understand the key terms or phrases used in this unit as listed below: Power series, Generating sequence, Generating functions, Diferrencing, Recurrence relation

Reflection

Think about other counting principle that can be applied to solve problems in Math and Computer.

4.1 GENERATING SEQUENCES AND FUNCTIONS

4.1.1 Generating series and sequence

We can define a generating function as a formal power series that represents a sequence of numbers or objects. A Power series is said to be infinite if it has infinite sum of terms of the form

$$\sum_{k=0}^{\infty} c_k x^k.$$

This can be expanded like $c_0 + c_1x + c_2x^2 + c_3x^3 + ...$ We call such power series in generating function as a generating series. Generating series generates the sequence $c_0, c_1, c_2...$ which is simply a sequence of coefficients of the infinite polynomial.

Example : What sequence is represented by generating series $3 + 8x^2 + x^3 +$

$\frac{x^5}{7} + 100x^6 ...$

Solution: $a_0 = 3$ since the coefficient of x^0 is 3. $a_1 = 0$ since coefficient of x^1=0, $a_2 = 8$ since x^2=8. Continuing we have $a_3 = 1$, $a_4 = \frac{1}{7}$, $a_5 = 100$.
Therefore the sequence is $3, 0, 8, 1, \frac{1}{7}, 100, ...$

Note: In generating function, we always start our sequence with a_0

Activity 2a

1. What sequence is represented by the following generating series $1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{4}{24} ... + \frac{x^n}{n!}$ + ...

4.2 Generating functions

4.2.1 Building generating Functions

We can build a closed formula for generating functions from power series using multiply, shift and subtract technique. Lets now discuss the following example.

Example: Find the generating function of the following generating series.

Solution:

$$\begin{aligned}
S &= 1 + x + x^2 + x^3 \\
-xS &= \quad x + x^2 + x^3 + ... \\
\hline
(1-x)S &= 1 \\
S &= \frac{1}{1-x} \\
\text{Therefore} \quad 1 + x + x^2 + x^3 &= \frac{1}{1-x}
\end{aligned}$$

Hence the generating function of 1, 1, 1, 1, ... is $\dfrac{1}{1-x}$

4.2.2 Differencing

Now that we know how to find generating functions for any constant sequence. Lets apply the concept differencing in the following example.

Example: Find the generating function of 1, 3, 5, 7,9 ...

Solution: Lets denote the generating function for 1, 3, 5, 7, 9, ... by A

We have

$$\begin{array}{rcl} A & = & 1+3x+5x^2+7x^3+9x^4 \\ -xA & = & \quad x+3x^2+5x^3+7x^4... \\ \hline (1-x)A & = & 1+2x+2x^2+2x^3+2x^4... \\ (1-x)A & = & 1+\dfrac{2x}{1-x} \\ A & = & \dfrac{1}{1-x}+\dfrac{2x}{(1-x)(1-x)} \\ A & = & \dfrac{1+x}{(1-x)^2} \end{array}$$

4.2.3 Solving recurrence relations with generating functions

Example: The sequence 1,3,7,15,31,63,... satisfies the recurrence relation $a_n = 3a_{n-1} - 2a_{n-2}$. Find the generating function of the sequence.

Solution : Given that

$$a_n = 3a_{n-1} - 2a_{n-2}$$

$$a_n - 3a_{n-1} + 2a_{n-2} = 0$$

$$1 - 3x + 2x^2 = 0$$

Lets call the generating function for the sequence by A. Then the following holds

$$
\begin{array}{rcl}
A & = & 1 + 3x + 7x^2 + 15x^3 + 31x^4 + ...a_n x^n + ... \\
-3xA & = & 0 - 3x - 9x^2 21x^3 - 45x^4 + ... - 3a_n x^n + ... \\
2x^2 A & = & 0 + 0x + 2x^2 + 6x^3 + 14x^4 + ... + 2a_n x^n + ... \\
\hline
(1 - 3x + 2x^2)A & = & 1 + 0 \\
A & = & \dfrac{1}{1 - 3x + 2x^2}
\end{array}
$$

4.3

Activity 4b

1. Compute the generating function of the following generating series $1 + 2x + 3x^2 + 4x^3$

4.4 Summary

In this Unit, We have learnt about Generating series, sequences and functions respectively. In the next unit, you will learn about Algorithms.

4.5 End of Unit Test

1. Find the generating function for 1, 4, 9, 16, ...hint: take 1 = a_0

2. Multiply the sequence 1,2,3,4... by the sequence 1,2,4,6, 8,...16, ...

3. Solve the recurrence relation given by $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 2$, with initial conditions $a_0 = 1$ and $a_1 = 2$.

Answers and Hints to Activities

Activity 4a

1. 1, 1, $\frac{1}{2}, \frac{1}{6}, \frac{1}{24}, \ldots \frac{1}{n!}$

Activity 4b

1. $\dfrac{1}{(1-x)^2}$

4.6 Further Reading

1. Epp, S.S. (2003).*Discrete Mathematics with Applications*, 4th Edition. Boston: Richard Stratton.

2. Lipschutz, S., Lipson, M. (2005).*Discrete Mathematics*. Tata Mc Graw Hill.

3. Rosen, K.H., (2012). *Discrete Mathematics and its Applications*, New York: McGraw Hill.

unit 5

ALGORITHMS

Introduction

There are many general classes of problems that arise in discrete mathematics. For instance: given a sequence of integers, find the largest one, given a network, find the shortest path between two vertices. When presented with such a problem, the first thing to do is to construct a model that translates the problem into a mathematical context. Setting up the appropriate mathematical model is only part of the solution.

To complete the solution, a method is needed that will solve the general problem using the model. Ideally, what is required is a procedure that follows a sequence of steps that leads to the desired answer. Such a sequence of steps is called an algorithm.Therefore in this unit, we will discuss algorithms that solve wide varieties of problems.

Intended Learning Outcomes

By the end of this Unit, you should be able to:

a. Use algorithms to solve different problems .

b. apply the time complexity of the algorithms in solving problems

c. solve problems relating to growth functions

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below: Algorithm, growth function, complexity

Reflection
Think about other mathematical models that can be applied to solve problems in Math and Computer.

5.1 ALGORITHM AND THEIR PROPERTIES

5.1.1 Algorithms

Definition 5.1. *An algorithm is a finite set of precise instructions for performing a computation or for solving a problem.*

Example1 : Describe an algorithm for finding the maximum (largest) value in a finite sequence of integers.

Solution

Method 1, we will simply use the English language to describe the sequence of steps used. We perform the following steps.

1. Set the temporary maximum equal to the first integer in the sequence. (The temporary maximum will be the largest integer examined at any stage of the procedure.)

2. Compare the next integer in the sequence to the temporary maximum, and if it is larger than the temporary maximum, set the temporary maximum equal to this integer.

3. Repeat the previous step if there are more integers in the sequence.

4. Stop when there are no integers left in the sequence. The temporary maximum at this point is the largest integer in the sequence.

Method 2: we can also use computer language using pseudo code description as follows

ALGORITHM 1 Finding the Maximum Element in a Finite Sequence.

procedure $\max(a_1, a_2, \ldots, a_n :$ integers $)$

$\max := a_1$

for $i := 2$ to n

  if $\max < a_i$ then $\max := a_i$

$\{\max$ is the largest element $\}$

5.1.2 Properties

There are several properties that algorithms generally share. They are useful to keep in mind when algorithms are described. These properties are:

- **Input**. An algorithm has input values from a specified set.

- **Output**. From each set of input values an algorithm produces output values from a specified set. The output values are the solution to the problem.

- **Definiteness**. The steps of an algorithm must be defined precisely.

- Correctness. An algorithm should produce the correct output values for each set of input values.

- **Finiteness**. An algorithm should produce the desired output after a finite (but perhaps large) number of steps for any input in the set.

- **Effectiveness**. It must be possible to perform each step of an algorithm exactly and in a finite amount of time.

- **Generality**. The procedure should be applicable for all problems of the desired form, not just for a particular set of input values.

5.1.3 Searching Algorithms

The general searching problem can be described as follows: Locate an element x in a list of distinct elements $a_1, a_2, \ldots, a_n$, or determine that it is not in the list. The solution to this search problem is the location of the term in the list that equals x (that is, i is the solution if $x = a_i$) and is 0 if x is not in the list.

Linear search

The linear search algorithm begins by comparing x and a_1. When $x = a_1$, the solution is the location of a_1, namely, 1 . When $x \neq a_1$, compare x with a_2. If $x = a_2$, the solution is the location of a_2, namely, 2. When $x \neq a_2$, compare x with a_3. Continue this process, comparing x successively with each term of the list until a match is found, where the solution is the location of that term, unless no match occurs. If the entire list has been searched without locating x, the solution is 0 . The pseudocode for the linear search algorithm is left as your exercise.

The Binary search

This second searching algorithm is called the binary search algorithm. It proceeds by comparing the element to be located to the middle term of the list. The list is then split into two smaller sublists of the same size, or where one of these smaller lists has one fewer term than the other. The search continues by restricting the search to the appropriate sublist based on the comparison of the element to be located and the middle term.

Example: Search for 19 in the list 1 2 3 5 6 7 8 10 12 13 15 16 18 19 20 22 .

Solution:

$$1\ 2\ 3\ 5\ 6\ 7\ 8\ 10 \qquad 12\ 13\ 15\ 16\ 18\ \ 19\ 20\ 22\ \&10 \quad < \quad 19$$

$$12\ 13\ 15\ 16 \qquad 18\ 1920\ 22\ \ \&16 \quad < \quad 19$$

$$18\ 19 \qquad 20\ 22$$

$$18\ 19$$

Therefore 19 is located as a 14^{th} term in a list.

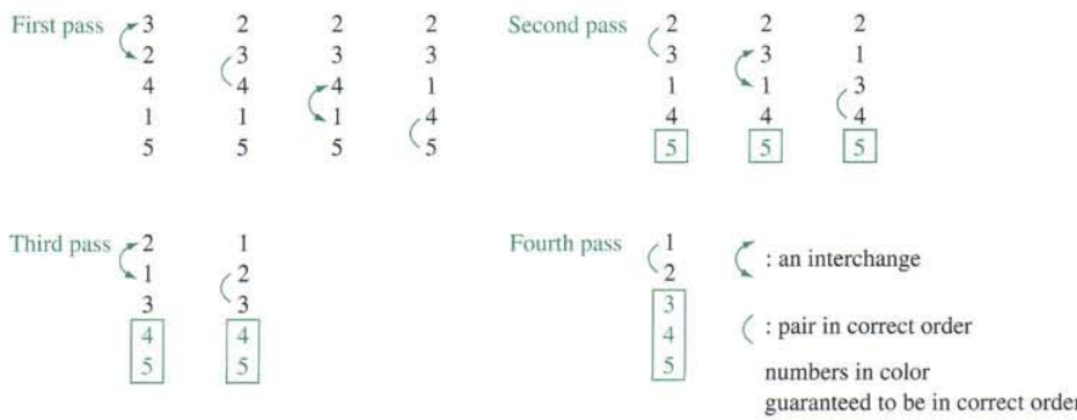
5.1.4 Sorting

Sorting is putting these elements into a list in which the elements are in increasing order. For instance, sorting the list $7, 2, 1, 4, 5, 9$ produces the list $1, 2, 4, 5, 7, 9$. Sorting the ist d, h, c, a, f (using alphabetical order) produces the list a, c, d, f, h.
There are two sorting algorithms, the bubble sort and he insertion sort.

Bubble sort

Example : Use the Bubble sort to put $3, 2, 4, 1, 5$ into increasing order

Solution : The Steps of a Bubble Sort are as follows:



We can also solve the above example using insertion sort algorithm

Insertion sort

ALGORITHM : The Insertion Sort.

procedure insertion $\mathrm{sort}\,(a_1, a_2, \ldots, a_n :$ real numbers with $n \geq 2)$

for $j := 2$ to n

begin

$i := 1$

while $a_j > a_i$

$i := i + 1$

$m := a_j$

for $k := 0$ to $j - i - 1$

$a_{j-k} := a_{j-k-1}$

$a_i := m$

end $\{a_1, a_2, \ldots, a_n$ are sorted $\}$

Example : Use the Insertion sort to put $3, 2, 4, 1, 5$ into increasing order

Solution:

$3,\ 2,\ 4,\ 1,\ 5$	The entire list.
$3 > 2$	Compare 2 and 3 from the list.
$2,\ 3,\ 4,\ 1,\ 5$	place 2 in first position of the list.
$4 > 2$ & $4 > 3$	4 placed in third position.
$1 < 2$	compare 1 and 2 from the sorted list
$1,\ 2,\ 3,\ 4,\ 5$	Correct order of the entire list.

5.1.5 Greedy Algorithm

Greedy algorithm is another type of algorithm that is also used to solve optimization problem. We now present an example of this algorithm.

Example : Consider the problem of making n cents change with quarters, dimes, nickels, and pennies, and using the least total number of coins. We can devise a greedy algorithm for making change for n cents by making a locally optimal choice at each step; that is, at each step we choose the coin of the largest denomination possible to add to the pile of change without exceeding n cents. For example, to make change for 67 cents, we first select a quarter (leaving 42 cents). We next select a second quarter (leaving 17 cents), followed by a dime (leaving 7 cents), followed by a nickel (leaving 2 cents), followed by a penny (leaving 1 cent), followed by a penny.

Activity 5a

1. Show all the steps used in binary insertion sort to sort the list 3,2, 4, 5, 1, 6

2. Describe an algorithm for finding the smallest integer in a finite sequence of natural numbers

5.2 Growth of functions and complexity of Algorithm

5.2.1 Growth of functions

The growth function is often described using the special Big O notation. This notation is used extensively to estimate the number of operations an algorithm uses as its input grows. We can therefore determine whether it is practical to use a particular algorithm to solve a problem as the size of the input increases.

Definition 5.2. *Let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. We say that $f(x)$ is $O(g(x))$ if there are constants C and k such that*

$$|f(x)| \leq C|g(x)|$$

whenever $x > k$. [This is read as " $f(x)$ is big-oh of $g(x)$."]

Example: Show that $7x^2$ is $O\left(x^3\right)$.

Solution: Note that when $x > 7$, we have $7x^2 < x^3$. (We can obtain this inequality by multiplying both sides of $x > 7$ by x^2.) Consequently, we can take $C = 1$ and $k = 7$ as witnesses to establish the relationship $7x^2$ is $O\left(x^3\right)$. Alternatively, when $x > 1$, we have $7x^2 < 7x^3$, so that $C = 7$ and $k = 1$ are also witnesses to the relationship $7x^2$ is $O\left(x^3\right)$.

Example: Show that n^2 is not $O(n)$.

Solution: To show that n^2 is not $O(n)$, we must show that no pair of constants C and k exist such that $n^2 \leq Cn$ whenever $n > k$. To see that there are no such constants, observe that when $n > 0$ we can divide both sides of the inequality $n^2 \leq Cn$ by n to obtain the equivalent inequality $n \leq C$. We now see that no matter what C and $\bar{k}$ are, the inequality $n \leq C$ cannot hold for all n with $n > k$. In particular, once we set a value of k, we see that when n is larger than the maximum of k and C, it is not true that $n \leq C$ even though $n > k$.

$NOTE$: There is a strong connection between big- O and big-Omega notation. In particular, $f(x)$ is $\Omega(g(x))$ if and only if $g(x)$ is $O(f(x))$.

Definition 5.3. *Let f and g be functions from the set of integers or the set of real numbers to the set of real numbers. We say that $f(x)$ is $\Omega(g(x))$ if there are positive constants C and k such that*

$$|f(x)| \geq C|g(x)|$$

whenever $x > k$. [This is read as " $f(x)$ is big-Omega of $g(x)$. "]

Example: The function $f(x) = 8x^3 + 5x^2 + 7$ is $\Omega(g(x))$, where $g(x)$ is the function $g(x) = x^3$. This is easy to see because $f(x) = 8x^3 + 5x^2 + 7 \geq 8x^3$ for all positive real numbers x. This is equivalent to saying that $g(x) = x^3$ is $O\left(8x^3 + 5x^2 + 7\right)$, which can be established directly by turning the inequality around.

5.2.2 Complexity of Algorithm

The time complexity of an algorithm can be expressed in terms of the number of operations used by the algorithm when the input has a particular size instead of actual computer time. This is because of the difference in time needed for different computers to perform basic operations. The operations used to measure time complexity can be the comparison of integers, the addition of integers, the multiplication of integers, the division of integers, or any other basic operation.

The fastest computers in existence can perform basic bit operations (for instance, adding, multiplying, comparing, or exchanging two bits) in 10^{-9} second (1 nanosecond), but personal computers may require 10^{-6} second (1 microsecond), which is 1000 times as long, to do the same operations. Next we will analyse the time complexity of some algorithms discussed in the previous section.
Example : Describe the time complexity of the linear search algorithm.

Solution: The number of comparisons used by the algorithm will be taken as the measure of the time complexity. At each step of the loop in the algorithm, two comparisons are performed one to see whether the end of the list has been reached and one to compare the element x with a term of the list. Finally, one more comparison is made outside the loop. Consequently, if $x = a_i, 2i + 1$ comparisons are used. The most comparisons, $2n + 2$, are required when the element is not in the list. In this case, $2n$ comparisons are used to determine that x is not a_i, for $i = 1, 2, \ldots, n$, an additional comparison is used to exit the loop, and one comparison is made outside the loop. So when x is not in the list, a total of $2n+2$ comparisons are used. Hence, a linear search requires at most $O(n)$ comparisons.

Example : Describe the time complexity of an Algorithm for finding the maximum element in a set.

Solution : The number of comparisons will be used as the measure of the time complexity of the algorithm, because comparisons are the basic operations used.

To find the maximum element of a set with n elements, listed in an arbitrary order, the temporary maximum is first set equal to the initial term in the list. Then, after a comparison has been done to determine that the end of the list has not yet been reached, the temporary maximum and second term are compared, updating the temporary maximum to the value of the second term if it is larger. This procedure is continued, using two additional comparisons for each term of the list-one to determine that the end of the list has not been reached and another to determine whether to update the temporary maximum. Because two comparisons are used for each of the second through the nth elements and one more comparison is used to exit the loop when $i = n + 1$, exactly $2(n - 1) + 1 = 2n - 1$ comparisons are used whenever this algorithm is applied. Hence, the algorithm for finding the maximum of a set of n elements has time complexity $\Theta(n)$, measured in terms of the number of comparisons used.

Activity 5b

1. Describe the time complexity of the binary search algorithm in terms of the number of comparisons used (and ignoring the time required to compute $m = [(i+j)/2]$ in each iteration of the loop in the algorithm).

2. Show that $x^5y^3 + x^4y^4 + x^3y^5$ is $\Omega\left(x^3y^3\right)$.

5.3 Summary

In this Unit, We have learnt about different Algorithms , growth functions and their time complexity. In the next unit, you will learn about Elementary number theory.

5.4 End of Unit Test

1. Show that Algorithm for finding the maximum element in a finite sequence of integers has all the properties listed.

2. 10. Show that x^3 is $O\left(x^4\right)$ but that x^4 is not $O\left(x^3\right)$.

3. Show that $3x^4 + 1$ is $O\left(x^4/2\right)$ and $x^4/2$ is $O\left(3x^4 + 1\right)$.

4, Show that $x \log x$ is $O\left(x^2\right)$ but that x^2 is not $O(x \log x)$.

5. Show that 2^n is $O\left(3^n\right)$ but that 3^n is not $O\left(2^n\right)$.

6. Explain what it means for a function to be $O(1)$.

7. Show that if $f(x)$ is $O(x)$, then $f(x)$ is $O\left(x^2\right)$.

8. Describe an algorithm that takes as input a list of n integers in nondecreasing order and produces the list of all values that occur more than once.

9. Describe an algorithm that takes as input a list of n integers and finds the number of negative integers in the list.

10 Describe an algorithm that takes as input a list of n integers and finds the location of the last even integer in the list or returns 0 if there are no even integers in the list.

11 How much time does an algorithm take to solve a problem of size n if this algorithm uses $2n^2 + 2^n$ bit operations, each requiring 10^{-9} second, with these values of n ?

[i]10

[ii.] 50.

5.5 Further Reading

1. Epp, S.S. (2003).$Discrete\ Mathematics\ with$ $Applications$, 4th Edition. Boston: Richard Stratton.

2. Lipschutz, S., Lipson, M. (2005).$Discrete Mathematics$. Tata Mc Graw Hill.

3. Rosen, K.H., (2012). *Discrete Mathematics and its Applications, NewYork : McGrawHill.*

unit 6

ELEMENTARY NUMBER THEORY

Introduction

In this unit, we will discuss some basic concepts of number theory including divisibility and modular arithmetic. We will cover prime numbers and begin the discussion of gcd. We will also describe some important algorithms such as the euclidean algorithm for computing the greatest common divisors.

Intended Learning Outcomes

By the end of this Unit, you should be able to:

a. solve problems involving prime numbers .

b. Find the greatest common divisor

c. Apply the Euclidean to find the GCD

d. Describe the Euler phi function

e. Solve Discrete log problem

f. Apply chinese remainder theorem in solving problem

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:
Prime numbers, Greatest Common Divisor, Euclidean algorithm .

Reflection
Think about other counting principle that can be applied to solve problems in Math and Computer.

6.1 INTEGERS AND PRIME NUMBERS

6.1.1 Integers

Definition 6.1. *If a and b are integers with $a \neq 0$, we say that a divides b if there is an integer c such that $b = ac$.*

When a divides b we say that a is a factor of b and that b is a multiple of a. The notation $a \mid b$ denotes that a divides b. We write $a \nmid b$ when a does not divide b.

Theorem 8. *Let a, b and c be integers. Then*

i. if $a \mid b$ and $a \mid c$, then $a \mid (b \pm c)$

ii. if $a \mid b$, then $a \mid bc$,for all integers c.

iii. if $a \mid b$ and $b \mid c$, then $a \mid c$

iv. if $a \mid b$ and $a \nmid c$, then $a \nmid (b \pm c)$

Proof. i. Suppose $a \mid b$ and $a \mid c$, by definition of divisibility $\exists$ s and t such that $b = as$ and $c = at$. Hence $b + c = as + at \Rightarrow a(s + t)$ or $b - c = as - at \Rightarrow a(s - t)$. Therefore a divides $a \mid (b \pm c)$. $\square$

Example: Determine whether $3 \mid 7$ and $3 \mid 12$

Solution: $3 \nmid 7$ because $7 \mid 3$ is not integer and $3 \mid 12$ because $12 \mid 3 = 4$

6.1.2 Division Algorithm

Let a be an integer and d a positive integer. Then there are unique integers q and r ,with $0 \leq r < d$, such that $a = dq + r$. where a = dividend, q = quotient, r =remainder, $d =$ is a divisor.

Note q= a div d and $r = a$ mod d

Example 3: What is the quotient and remainder when 101 is divided by 11?

Solution

$$\begin{aligned} a &= dq + r \\ 101 &= 11.9 + 2 \end{aligned}$$

The quotient when 101 is divided by 11 is 9= 101 div 11.

The remainder is 2 = 101 mod 11 .

6.1.3 Prime numbers

Definition 6.2. *Apositive integer p greater than 1 is called prime if the only positive factors of p are 1 and p. A positive integer that is greater than 1 and is not prime is called Composite.*

Example: (1) The integer 7 is prime because its only positive factors are 1 and 7 where as the integer 9 is composite because it is divisible by 3.
(2) Primes less than 50 are as follows: 2 , 3 , 5 , 7 , 11 , 13 , 17 , 19, 23 , 29 , 31, 37 ,41 ,43 ,47.
(3) Although 21, 24 , 17 and 29 are not primes, each can be written as a product of primes. e.g 21= 3.7 and 1729= 7.13.19.

Theorem 9. *Every integer $n > 1$ can be written as a product of primes.*

Theorem 10. *There is no largest prime, that is , $\exists$ an infinite number of primes.*

6.1.4 Factoring

Theorem 11 (The Fundamental Theorem of Arithmetic)**.** *Every positive integer greater than 1 can be written uniquely as a prime or product of two or more primes where the prime factors are written in order of increasing size.*

We are now going to look on examples that gives some prime factorization of integers.
Example: Write the prime factorization of the following integers 100, 999 ,641 and 1024

Solution : The prime factorization are as follows

$$\begin{aligned} 100 &= 2.2.5.5 = 2^2.5^2 \\ 999 &= 3.3.3.37 = 3^3.37 \\ 1024 &= 2.2.2.2.2.2.2.2.2.2 = 2^{10} \\ 641 &= 641 \end{aligned}$$

6.1.5 Testing for Primarity

We can test for primarity by following these two procedures:

(a) By observing if the if the integer is not divisible by any prime less than or equal to its square root.
(2) By trial division procedure.

Theorem 12. *If n is a composite integer , then n has a prime divisor less than or equal to* $\sqrt{n}$.

Example: Show that 101 is prime.
Solution: The only primes not exceeding $\sqrt{101}$ are 2, 3, 5 and 7. Because 101 is not divisible by 2, 3, 5 and 7 (the quotient of 101 and each of these integers is not an integer) . It follows that 101 is Prime.

In the following example we will show how trial division procedure is applied to test primarity.
Example : Find the prime factorization of 7007.

Solution : To find the prime factorization of 7007, first perform divisions of 7007 by successive primes, beginning with 2. None of the primes 2, 3 and 5 divides 7007

, However 7 divides 7007 with 7007 /7=1001. Next divide 1001 by successive prime beginning with 7. It is immediately seen that 7 also divides 1001 because 1001 /7=143. Continue dividing 143 by successive primes , we see 7 does not divide 143. 11 does divide 143 , and 143 /11= 13. So because 13 is prime the procedure is completed. Therefore the prime factorization of 7007 is 7 .7.11.13 = 7^2. 11. 13

Activity 2a

1. Determine whether each of these integers is prime

   (a) i. 21 ii. 71 iii. 97 and iv. 143

   (b)i. 19 ii. 107 iii. 101 and iv. 27

2. Find the prime factorization of each of these integers

   (a) 126

   (b) 1001

   (c)729

6.2 MODULAR ARITHMETIC AND EUCLIDEAN ALGORITHM

6.2.1 Modular Arithmetic

Definition 6.3. *If a and b are integers and m is a positive integer , then a is congruent b modulo m if m divides $a - b$. We use notation $a \equiv b$ (mod m) to indicate that a is congruent to b modulo m. If a and b are not congruent modulo m , we write $a \not\equiv b$ mod m.*

Theorem 13. *Let a and b be integers and let m bea positive integer. Then $a \equiv b$ (mod m) if and only if a mod m= b mod m.*

Example Determine whether

a. 17 is congruent to 5 modulo 6.

b. 24 is congruent to 14 modulo 6.

Solution

a. 17 is congruent to 5 modulo 6.
   Because 6 divides 17-5=12, we see that $17 \equiv 5$(mod) 6

b. 24 is congruent to 14 modulo 6.
   Because 24-14=10 is not divisible by 6, therefore 24 $\not\equiv$ 14 mod 6

Lets now looks on some properties of Mod. Let $m \in \mathbb{Z} \geq$2 with $a, b, c \in \mathbb{Z}$
$a \equiv a$ mod m
If $a \equiv b$ mod m then $b \equiv a$ mod m
If $a \equiv b$ mod m and $b \equiv c$ mod m then $a \equiv c$ mod m
If $a \equiv b$ mod m and $c \equiv d$ mod m then $a \pm c \equiv b\pm$ mod m and $a \times c \equiv b \times d$ mod m.

6.2.2 Greatest Common Divisor (GCD)

The largest integer that divides both of the two integers is called the greatest common divisor.

Definition 6.4. *Let a and b be integers not both zero.The largest integer d such that d$|$*

a and $d|\,b$ is called the greatest common divisor of a and b denoted by gcd(a , b)

Example What is the gcd of

a. 24 and 36

b. 17 and 22

Solution

a. 24 and 36
   Positive common divisors of 24 and 36 are 1, 2, 3, 4, 6 and 12. Therefore the gcd (24 , 36) = 12

b. The integers 17 and 22 have no positive common divisors other than 1. So the gcd (17, 22) =1

Definition 6.5. *The integers a and b are relatively prime if their greatest common divisor is 1. It follows that the integers 17 and 22 are relative prime because gcd (17, 22)= 1 .*

6.2.3 Euclidean Algorithm

This is where you repeatedly use division algorithm. At each step the previous divisor and remainder becomes the new dividend and divisor respectively.

Example: Find the gcd of 414 and 662 using the euclidean algorithm.

Solution: The successive uses of the division algorithm give

$$\begin{aligned} a &= dq + r \\ 662 &= 414 \times 1 + 248 \\ 248 &= 166 \times 1 + 82 \\ 166 &= 82 \times 2 + \underline{2} \\ 82 &= 2 \times 41 + 0 \end{aligned}$$

Hence the gcd(414, 662) is 2 because 2 is the last nonzero remainder.

Definition 6.6. *An element* $x \in \mathbb{Z}/m_z$ *has a multiplicative inverse* $\frac{1}{x}$ *or* x^{-1} *in* $\mathbb{Z}/m_z$ *when the* $\gcd(x, m) = 1$

In the next example we are going to compute the inverse using Euclidean Algorithm.
Example: Find the gcd of 11 mod 14 using the euclidean algorithm.
Solution:

$$
\begin{aligned}
&= 11 \; mod 14 \\
14 &= 11 \times 1 + 3 \\
11 &= 3 \times 3 + 2 \\
3 &= 2 \times 1 + \underline{1} \text{ Hence} \\
1 &= 3 - 2 \times 1 \\
1 &= 3 - [11 - 3 \times 3] \times 1 \\
1 &= 3 - 1[11 - 3 \times 3] \\
1 &= 3 - 1 \times 11 + 3 \times 3 \\
1 &= 4 \times 3 - 11 \times 1 \\
1 &= 4 \times 3 - 1 \times 11 \\
1 &= 4 \times [14 - 1 \times 11] - 1 \times 11 \\
1 &= 4 \times 14 - 4 \times 11 - 1 \times 11 \\
1 &= 4 \times 14 - 5 \times 11 \text{ mod} 14 \\
1 &= -5 \times 11 \text{ mod} 14 \\
&\equiv 9 \times 11 \text{mod} 14
\end{aligned}
$$

Therefore the inverse of 11 mod 14 is 9

6.3 EULER PHI (ϕ) FUNCTION

Let $n \in \mathbb{Z}_{>0}$, Let $\mathbb{Z}/n\mathbb{Z}^*$= $\{a \mid 1 \leq a \leq n, \gcd(a, n) = 1\}$. Then the Euler Phi function is denoted by $\phi(n)$ is | $\mathbb{Z}/n\mathbb{Z}^*$ |.

Example : $\mathbb{Z}/12\mathbb{Z}^*$= {1, 5, 7 , 11} . Therefore $\phi(12)$= 4 . Similarly we have $\mathbb{Z}/5\mathbb{Z}^*$= $\{1, 2, 3, 4\}$ with $\phi(5)$= 4 and $\phi(6)$= 2.

In general if P is Prime then $\phi(p)$= $p - 1$. If $r \geq 1$ and P is prime then $\phi(p^r) = p^r - p^{r-1}$ = $p^{r-1}(p - 1)$.

Example: Compute $\phi(5^3)$

Solution

$$\begin{aligned}
\phi(5^3) &= 5^{3-1}(5-1) \\
&= 5^2(5-1) \\
&= 5^2 \times 4 \\
&= 100
\end{aligned}$$

Theorem 14. *(Euler-Fermat Theorem)*

The theorem states that if n and a are co-prime positive integers such that $\phi(n)$ is an euler totient function then a raised to the power $\phi(n)$ is congruent to 1 modulo n.

That is , if the $\gcd(a, n) = 1$ then $a^{\phi(n)} = 1 \mod m$.

Example: Consider $\phi(10) = \phi(5)\ \phi(2) = 4 \times 1$, $\mathbb{Z}/10\mathbb{Z}^* = \{1, 3, 7, 9\}$. It is guaranteed that $1^{\phi(10)} \equiv 1^4 \equiv 1 \mod 10,\ 3^4 \equiv 1 \mod 10,\ 7^4 \equiv 1 \mod 10,$
$9^4 \equiv 1 \mod 10$

If $\gcd(c, m)$ and $a \equiv b \mod \phi(m)$ with $a,\ b \in \mathbb{Z} > 0$ then $C^a \equiv C^b \mod m$.

Example: Reduce $2^{3005} \mod 21$

Solution: We begin by computing $\phi(21)$

$$\begin{aligned}
\phi(21) &= \phi(7)\phi(3) \\
&= 6 \times 2 = 12 \\
3005 &\equiv 5 \mod 12 \quad \text{we take exponent modulo phi of 21}
\end{aligned}$$

Therefore $2^{3005} \equiv 2^5 \equiv 32 \equiv 11 \mod 21$.

If $\gcd(m,n)$ =1 then $\phi(mn)=\phi(m).\phi(n)$.

Example: Compute $\phi(720)$

Solution:

$$\begin{aligned}
\phi(720) &= \phi(2^4)\phi(3^2)\phi(5) \\
&= 2^{4-1}(2-1)3^{2-1}(3-1)(5-1) \\
&= 2^3(2-1)3^1(3-1)(5-1) \\
&= 2^3 \times 6 \times 4 \\
&= 48 \times 4 \\
&= 192
\end{aligned}$$

Theorem 15. *(Fermat's Little Theorem)*

The theorem states (i) If P is prime and $a \in \mathbb{Z}$ then $a^p \equiv a \mod p$. (ii.) If P does not divide a then $a^{p-1} \equiv 1 \mod p$.

Example : So it is guaranteed from part (i) of the theorem that $4^{11} \equiv 4 \mod 11$ Since 11 is prime.Also we have $6^{11} \equiv 6 \mod 11$. By part (ii) of the theorem we have $2^{10} \equiv 1 \mod 11$

Activity 2b

1. Find the Euler phi function of $\phi(296)$

2. Reduce $7^{222} \mod 10$

3. Compute $\phi(11^6)$

6.4 Discrete Logarithm Problem & Chinese Remainder Theorem

6.4.1 Discrete Logarithm Problem

Definition 6.7. *The discrete log of β to the base α is denoted as $log_{\alpha}\beta$ is a unique value integer x, $0 \leq x \leq n-1$ such that $\beta = \alpha^x$.*

Since $\beta = \alpha^x$ it implies that $x = log_{\alpha}\beta$. For Example 8=2^x, which implies that $x = log_2 8$. Introducing the modular Arithmetic , we get

$$\beta \equiv \alpha^x mod P \text{ implies } x \quad \rightarrow \quad log_{\alpha,P}(B)$$

Example: Express $26067 \equiv 14^{12}$ mod 29867 as a discrete log.

Solution: 12=$\log_{14,29867}(26067)$

Example: Given that $P = 19$, $Z_{19}^{*} = \ 1,\ 2,\ ...\ 18$. Generator $\alpha = 2$. Solve $\log_2 14$

Solution: Since $\alpha^x \equiv \ \beta \mod P$

x	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
α	2	4	8	16	13	7	14	9	18	17	15	11	3	3	6	12	10	1

$$2^x \equiv 14 \mod 19$$

$$2^7 \equiv 14 \mod 19$$

Therefore $\log_2 14 = 7$ in Z_{19}^*

6.4.2 CHINESE REMAINDER THEOREM

Theorem 16. *Let $m_1, m_2, m_3 ... m_r$ be a collection of a pair wise relatively prime integers. Then the system of congruences*

$$x \equiv a_1 \mod m_1$$
$$x \equiv a_2 \mod m_2$$
$$x \equiv a_3 \mod m_3$$
$$\vdots$$
$$x \equiv a_r \mod M_r$$

has a solution modulo $M = m_1, m_2, m_3, ...m_r$ for the integers $a_1, a_2, ..., a_r$.

Proof: To establish this theorem, we need to show that a solution exists and that it is unique modulo m. We will show that a solution exists by describing a way to construct this solution; showing that the solution is unique modulo m is left as exercise at the end of this section. To construct a simultaneous solution, first let

$$M_k = m/m_k$$

for $k = 1, 2, \ldots, n$. That is, M_k is the product of the moduli except for m_k. Because m_i and m_k have no common factors greater than 1 when $i \neq k$, it follows that $\gcd(m_k, M_k) = 1$. Consequently, by Theorem 3 , we know that there is an integer y_k, an inverse of M_k modulo m_k, such that

$$M_k y_k \equiv 1 \,(\mathrm{mod} m_k)$$

To construct a simultaneous solution, form the sum

$$x = a_1M_1y_1 + a_2M_2y_2 + \cdots + a_nM_ny_n.$$

We will now show that x is a simultaneous solution. First, note that because $M_j \equiv 0\,(\mathrm{mod} m_k)$ whenever $j \neq k$, all terms except the k th term in this sum are congruent to 0 modulo m_k. Because $M_ky_k \equiv 1\,(\mathrm{mod} m_k)$ we see that

$$x \equiv a_kM_ky_k \equiv a_k\,(\mathrm{mod} m_k)\,,$$

for $k = 1, 2, \ldots, n$. We have shown that x is a simultaneous solution to the n congruences.

The following example illustrates how to use the construction given in the proof of Theorem to solve a system of congruences.

Example What are the solutions of the systems of congruences

$$x \equiv 2(\mathrm{mod} 3)$$
$$x \equiv 3(\mathrm{mod} 5)$$
$$x \equiv 2(\mathrm{mod} 7)?$$

Solution To solve the system of congruences in Example 5, first let $m = 3 \cdot 5 \cdot 7 = 105, M_1 = m/3 = 35$, $M_2 = m/5 = 21$, and $M_3 = m/7 = 15$. We see that 2 is an inverse of $M_1 = 35$ modulo 3 , because $35 \equiv 2(\mathrm{mod} 3); 1$ is an inverse of $M_2 = 21\,\mathrm{modulo}\,5$, because $21 \equiv 1(\mathrm{mod} 5)$; and 1 is an inverse of $M_3 = 15(\mathrm{mod} 7)$, because $15 \equiv 1(\mathrm{mod} 7)$. The solutions to this system are those x such that

$$\begin{aligned} x \equiv a_1M_1y_1 + a_2M_2y_2 + a_3M_3y_3 &= 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 \\ &= 233 \equiv 23(\mathrm{mod} 105). \end{aligned}$$

It follows that 23 is the smallest positive integer that is a simultaneous solution. We conclude that 23 is the smallest positive integer that leaves a remainder of 2 when divided by 3 , a remainder of 3 when divided by 5 , and a remainder of 2 when divided by 7 .

6.5 Summary

In this Unit, We have learnt about elementary number theory. We have discussed on how to solve problems using Euclidean algorithm, Chinese remainder theorem, Discrete logarithm and Euler phi functions. In the next unit, you will learn about Graph theory.

6.6 End of Unit Test

1. Express the greatest common divisor of each of these pairs of integers as a linear combination of these integers. a) 10,11

   b) 21,44

   c) 36,48

   d) 117,213

2. Express the greatest common divisor of each of these pairs of integers as a linear combination of these integers. a) 9,11

   b) 33,44

   c) 35,78

   e) 101,203

3. Show that 15 is an inverse of 7 modulo 26 .

4. Show that 937 is an inverse of 13 modulo 2436 .

5. Find an inverse of 4 modulo 9.

6. Find an inverse of 2 modulo 17 .

7. Find an inverse of 19 modulo 141 .

8. Find an inverse of 144 modulo 233.

9. Find all solutions to the system of congruences.

$$x \equiv 2(\mathrm{mod} 3)$$
$$x \equiv 1(\mathrm{mod} 4)$$
$$x \equiv 3(\mathrm{mod} 5)$$

10. Find all solutions to the system of congruences.

$$x \equiv 1(\mathrm{mod} 2)$$
$$x \equiv 2(\mathrm{mod} 3)$$
$$x \equiv 3(\mathrm{mod} 5)$$
$$x \equiv 4(\mathrm{mod} 11)$$

11. Determine whether each of these integers is prime. a) 19
    b) 27
    c) 93
    d) 101
    e) 107
    f) 113

12. Find the prime factorization of each of these integers. a) 88

b) 126

c) 729

d) 1001

e) 1111

f) 909,090

13. Find the prime factorization of each of these integers. a) 39

    b) 81

    c) 101

    d) 143

    e) 289

    f) 899

14 Use the Euclidean algorithm to find a) $\gcd(12, 18)$.

b) $\gcd(111, 201)$.

c) $\gcd(1001, 1331)$.

d) $\gcd(12345, 54321)$.

e) $\gcd(1000, 5040)$.

f) $\gcd(9888, 6060)$.

15. Use the Euclidean algorithm to find a) $\gcd(1, 5)$.

    b) $\gcd(100, 101)$.

    c) $\gcd(123, 277)$.

    d) $\gcd(1529, 14039)$.

    e) $\gcd(1529, 14038)$.

    f) $\gcd(11111, 111111)$.

6.7 Further Reading

1. Epp, S.S. (2003).*Discrete Mathematics with Applications*, 4th Edition. Boston: Richard Stratton.

2. Lipschutz, S., Lipson, M. (2005). *Discrete Mathematics*. Tata Mc Graw Hill.

3. Rosen, K.H., (2012). *Discrete Mathematics and its Applications*, New York: McGraw Hill.

unit 7

GRAPH THEORY

Introduction

Graphs are discrete structures consisting of vertices and edges that connect these vertices. Problems in almost every conceivable discipline can be solved using graph models. We will give examples to show how graphs are used as models in a variety of areas. For instance, we will show how graphs are used to represent the competition of different species in an ecological niche, how graphs are used to represent who influences whom in an organization, and how graphs are used to represent the outcomes of round-robin tournaments.

Therefore in this unit will introduce the basic concepts of graph theory and present many different graph models. To solve the wide variety of problems that can be studied using graphs, we will introduce many different graph algorithms.

Intended Learning Outcomes

By the end of this Unit, you should be able to:

a. Define a graph .

b. Identify even and odd degree vertices

c. Describe the path, connectivity, cycle and trees

d. Distinguish between Eulerian and Halmitonian path.

e. Find the optimum- best tree

Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below:
graphs, connectivity, path and trees .

Reflection :
Think about networks, communication links between computers, molecular formulars and e.t.c. What comes in your mind?

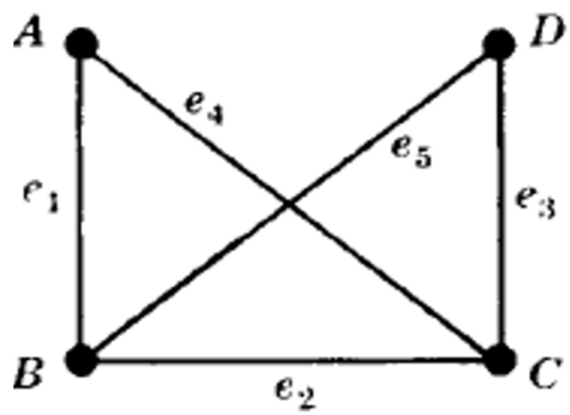
7.1 Basic concepts on a graph

7.1.1 Undirected graphs

Definition 7.1 (Undirected graphs). *A graph G consists of two things: - A set $V = V(G)$ whose elements are called vertices, points, or nodes of G. - A set $E = E(G)$ of unordered pairs of distinct vertices called edges of G.*

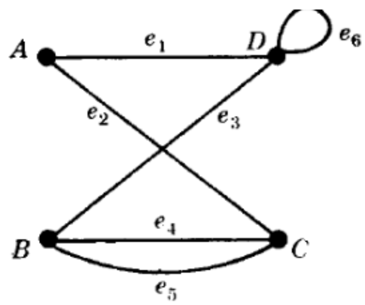
We denote such a graph by $G(V, E)$ or equivalently by $G = (V, E)$ when we want to emphasize the two parts of G.

A graph in which each edge connects two different vertices and where no two edges connect the same pair of vertices is called **a simple graph**.



In a simple graph, each edge is associated to an unordered pair of vertices, $\{u, v\}$, and no other edge is associated to this same edge, that is to say we have $\{u, v\}$ as an edge of the graph.

Graphs that may have multiple edges connecting the same vertices are called $Multigraphs.$ When there are m different edges associated to the same unordered pair of vertices $\{u, v\}$, we also say that $\{u, v\}$ is an edge of multiplicity m. That is, we can think of this set of edges as m different copies of an edge $\{u, v\}$.



(b) Multigraph

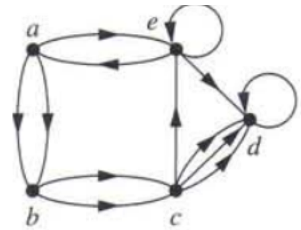
Sometimes a communications link connects a data center with itself, perhaps a feedback loop for diagnostic purposes. To model this network we need to include edges that connect a vertex to itself. Such edges are called loops. Graphs that may include loops, and possibly multiple edges connecting the same pair of vertices, are sometimes called $Pseudographs$.

7.1.2 Directed graphs

Definition 7.2. *A directed graph (or digraph) (V, E) consists of a nonempty set of vertices V and a set of directed edges (or arcs) E. Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to start at u and end at v.*

When we depict a directed graph with a line drawing, we use an arrow pointing from u to v to indicate the direction of an edge that starts at u and ends at v. A directed graph may contain loops and it may contain multiple directed edges that start and end at the same vertices.

A directed graph may also contain directed edges that connect vertices u and v in both directions; that is, when a digraph contains an edge from u to v, it may also contain one or more edges from v to u. Note that we obtain a directed graph when we assign a direction to each edge in an undirected graph. Below is an example of a directed graph



7.1.3 Basic graph teminology and special types

First, we give some terminology that describes the vertices and edges of undirected graphs.

Definition 7.3. *Two vertices u and v in an undirected graph G are called adjacent (or neighbors) in G if u and v are endpoints of an edge of G. If e is associated with $\{u, v\}$, the edge e is called incident with the vertices u and v. The edge e is also said to connect u and v. The vertices u and v are called endpoints of an edge associated with $\{u, v\}$.*

To keep track of how many edges are incident to a vertex, we make the following definition.

Definition 7.4. *The degree of a vertex in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.*

Example: What are the degrees of the vertices in the graphs G and H displayed in Figure 1 ?

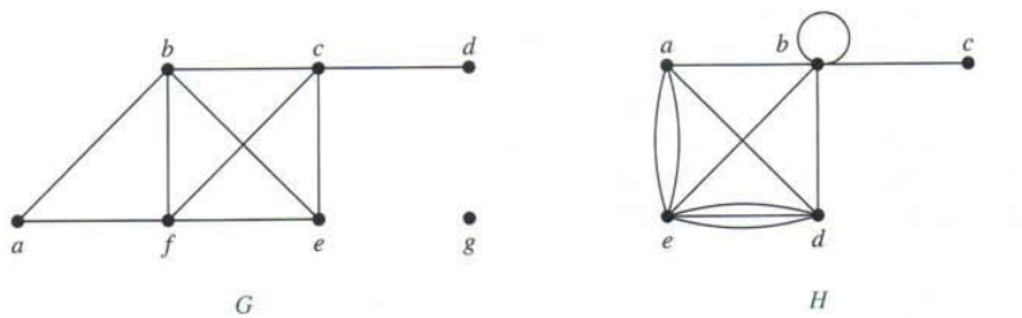


FIGURE 1 **The Undirected Graphs G and H.**

Solution: In $G, \deg(a) = 2, \deg(b) = \deg(c) = \deg(f) = 4, \deg(d) = 1, \deg(e) = 3$, and $\deg(g) = 0$. In $H, \deg(a) = 4, \deg(b) = \deg(e) = 6, \deg(c) = 1$, and $\deg(d) = 5$

Theorem 17 (THE HANDSHAKING THEOREM). *Let $G = (V, E)$ be an undirected graph with e edges. Then*

$$2e = \sum_{v \in V} \deg(v).$$

Note that this theorem applies even if multiple edges and loops are present. **Example**: How many edges are there in a graph with 10 vertices each of degree six?

Solution: Because the sum of the degrees of the vertices is $6 \cdot 10 = 60$, it follows that $2e = 60$. Therefore, $e = 30$.

Theorem 16 shows that the sum of the degrees of the vertices of an undirected graph is even. This simple fact has many consequences, one of which is given as Theorem 17.

Theorem 18. *An undirected graph has an even number of vertices of odd degree.*

Proof: Let V_1 and V_2 be the set of vertices of even degree and the set of vertices of odd degree, respectively, in an undirected graph $G = (V, E)$.

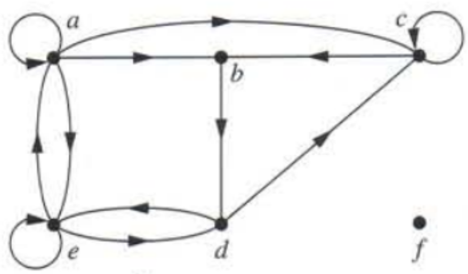
$$2e = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$$

Because $\deg(v)$ is even for $v \in V_1$, the first term in the right-hand side of the last equality is even. Furthermore, the sum of the two terms on the right-hand side of the last equality is even, because this sum is $2e$. Hence, the second term in the sum is also even. Because all the terms in this sum are odd, there must be an even number of such terms. Thus, there are an even number of vertices of odd degree.

Definition 7.5. *When (u, v) is an edge of the graph G with directed edges, u is said to be adjacent to v and v is said to be adjacent from u. The vertex u is called the initial vertex of (u, v), and v is called the terminal or end vertex of (u, v). The initial vertex and terminal vertex of a loop are the same.*

Definition 7.6. *In a graph with directed edges the in-degree of a vertex v, denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex. The out-degree of v, denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex. (Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.)*

Example Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in Figure below



G

Solution: The in-degrees in G are $\deg^-(a) = 2, \deg^-(b) = 2, \deg^-(c) = 3, \deg^-(d) = 2$, $\deg^-(e) = 3$, and $\deg^-(f) = 0$. The out-degrees are $\deg^+(a) = 4, \deg^+(b) = 1, \deg^+(c) = 2$, $\deg^+(d) = 2, \deg^+(e) = 3$, and $\deg^+(f) = 0$.

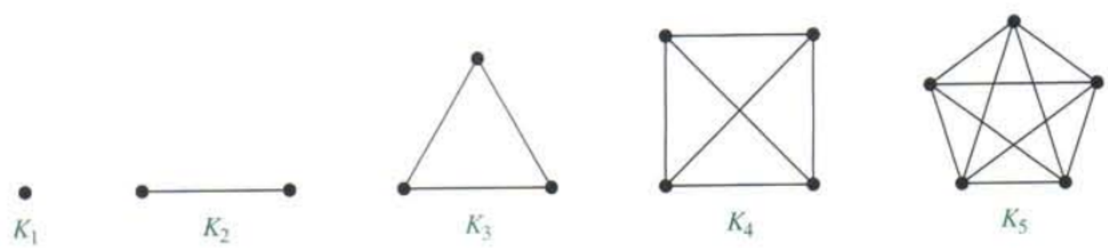
Theorem 19. *Let $G = (V, E)$ be a graph with directed edges. Then*

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|.$$

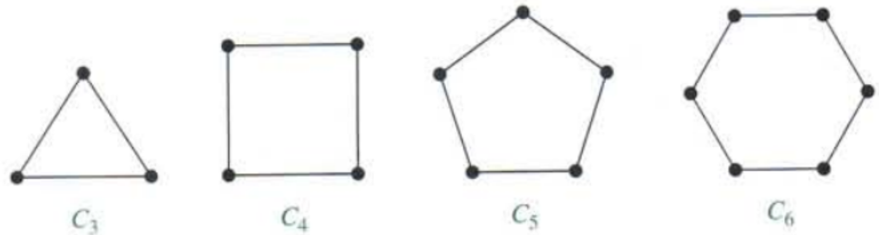
Special graphs

Let now discuss some of special graphs.

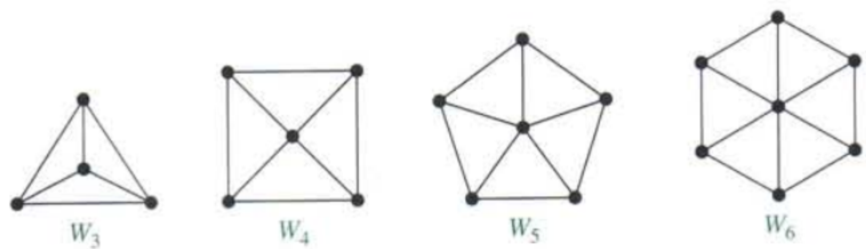
$\mathbf{1}$.$\mathbf{Complete graph}$: $\mathrm{K_n}$, is the simple graph that contains exactly one edge between each pair of distinct vertices. Representation Example: $\mathrm{K_1},\ \mathrm{K_2},\ \mathrm{K_3},\ \mathrm{K_4},\ \mathrm{K_5}$



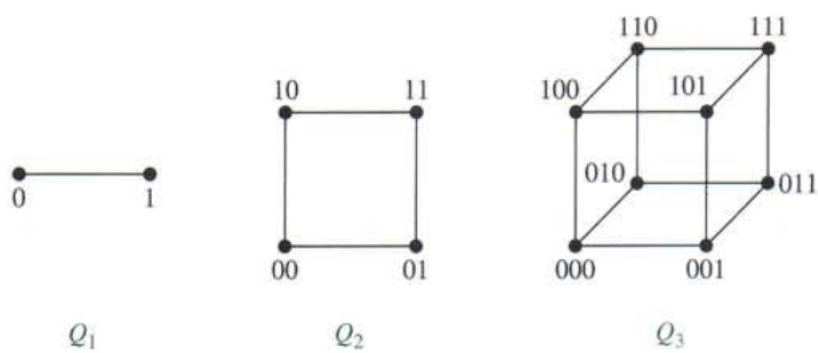
$\mathbf{2}$.$\mathbf{Cycles}$. The cycle $C_n, n \geq 3$, consists of n vertices $v_1, v_2, \ldots, v_n$ and edges $\{v_1,$ $\{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$. The cycles C_3, C_4, C_5, and C_6 are displayed Figure below



3. **Wheels**. We obtain the wheel W_n when we add an additional vertex to the cycle C_n, for n and connect this new vertex to each of the n vertices in C_n, by new edges. The wheels W_3, W_5, and W_6 are displayed in Figure below.



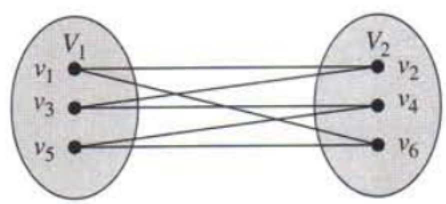
4. **n – Cubes**. The ***n***-dimensional hypercube, or ***n***-cube, denoted by Q_n , is the graph that has vertices representing the 2^n bit strings of length n. Two vertices are adjacent if and only if the bit strings that they represent differ in exactly one bit position. The graphs Q_1, Q_2, and Q_3 are displayed in Figure below.



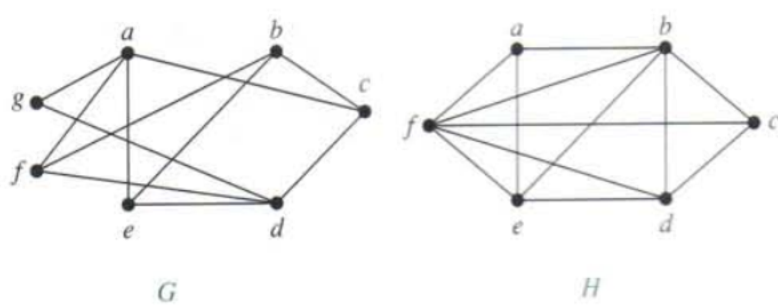
BIPARTITE GRAPHS.

Definition 7.7. *A simple graph G is called bipartite if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a bipartition of the vertex set V of G.*

Example The figure below is C_6 which is bipartite, because its vertex set can be partitioned into the two sets $V_1 = \{v_1, v_3, v_5\}$ and $V_2 = \{v_2, v_4, v_6\}$, and every edge of C_6 connects a vertex in V_1 and a vertex in V_2.



Example: Are the graphs G and H displayed in Figure below bipartite?



Solution: Graph G is bipartite because its vertex set is the union of two disjoint sets, $\{a, b, d\}$ and $\{c, e, f, g\}$, and each edge connects a vertex in one of these subsets to a vertex in the other subset. (Note that for G to be bipartite it is not necessary that every vertex in $\{a, b, d\}$ be adjacent to every vertex in $\{c, e, f, g\}$. For instance, b and g are not adjacent.)

Graph H is not bipartite because its vertex set cannot be partitioned into two subsets so that edges do not connect two vertices from the same subset. (You should verify this by considering the vertices a, b, and f.)

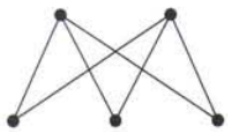
Theorem 20. *A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.*

The proof for theorem 19 is left for you as an exercise.

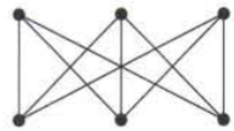
Complete Bipartite graphs :

Definition 7.8. *The complete bipartite graph $K_{m,n}$ is the graph that has its vertex set partitioned into two subsets of m and n vertices, respectively. There is an edge between two vertices if and only if one vertex is in the first subset and the other vertex is in the second subset.*

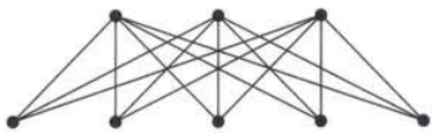
The complete bipartite graphs $K_{2,3}, K_{3,3}, K_{3,5}$, and $K_{2,6}$ are displayed in Figure below.



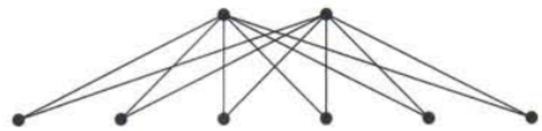
$K_{2,3}$



$K_{3,3}$



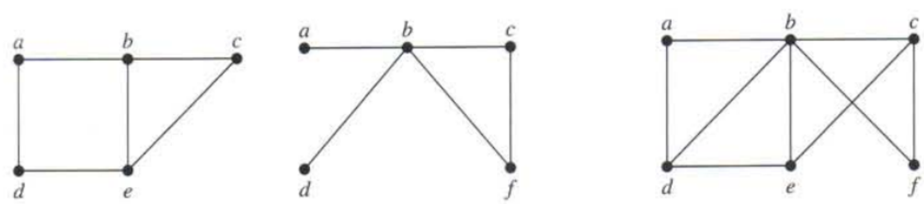
$K_{3,5}$



$K_{2,6}$

Definition 7.9 (Subgraphs)**.** *A subgraph of a graph $G = (V, E)$ is a graph $H = (W, F)$, where $W \subseteq V$ and $F \subseteq E$. A subgraph H of G is a proper subgraph of G if $H \neq G$.*

The union of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the simple graph with vertex set $V_1 \cup V_2$ and edge set $E_1 \cup E_2$. The union of G_1 and G_2 is denoted by $G_1 \cup G_2$. Let look on the example of a subgraph.



G_1 G_2 $G_1 \cup G_2$

(a) (b)

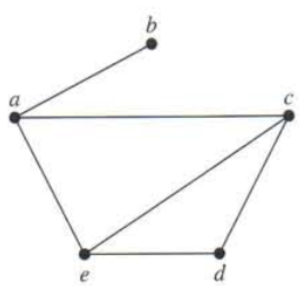
7.1.4 Graph representation and Isomorphism

Adjacency and incidence list/matrices

One way to represent a graph without multiple edges is to list all the edges of this graph. Another way to represent a graph with no multiple edges is to use adjacency lists, which specify the vertices that are adjacent to each vertex of the graph.
Incidence(Matrix) : Most useful when information about edges is more desirable than information about vertices.

Adjacency(Matrix/List) : Most useful when information about the vertices is more desirable than information about the edges.

Example The table below is the adjacency list that describe a simple graph



A Simple Graph.

TABLE 1 An Adjacency List for a Simple Graph.

Vertex	*Adjacent Vertices*
a	b, c, e
b	a
c	a, d, e
d	c, e
e	a, c, d

Example Represent the directed graph shown in Figure by listing all the vertices that are the terminal vertices of edges starting at each vertex of the graph.

Solution: Table represents the directed graph shown in Figure 2.

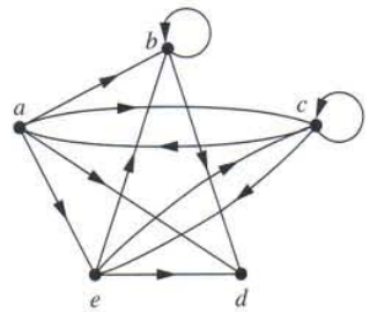


FIGURE A Directed Graph.

TABLE 2 An Adjacency List for a Directed Graph.

Initial Vertex	*Terminal Vertices*
a	b, c, d, e
b	b, d
c	a, c, e
d	
e	b, c, d

Definition 7.10. *Suppose that $G = (V, E)$ is a simple graph where $|V| = n$. Suppose that the vertices of G are listed arbitrarily as $v_1, v_2, \ldots, v_n$. The adjacency matrix $\mathbf{A}$ (or $\mathbf{A}_G$) of G, with respect to this listing of the vertices, is the $n \times n$ zero-one matrix with 1 as its (i, j) th entry when v_i and v_j are adjacent, and 0 as its (i, j) th entry when they are not adjacent. In other words, if its adjacency matrix is $\mathbf{A} = [a_{ij}]$, then*

$$a_{ij} = \begin{cases} 1 & \textit{if } \{v_i, v_j\} \textit{ is an edge of } G, \\ 0 & \textit{otherwise.} \end{cases}$$

Example: Use an adjacency matrix to represent the graph shown in Figure .

Simple Graph.

Solution: We order the vertices as a, b, c, d. The matrix representing this graph is

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

Example :Draw a graph with the adjacency matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

with respect to the ordering of vertices a, b, c, d.

Solution: A graph with this adjacency matrix is shown in Figure below.

Another common way to represent graphs is to use incidence matrices.

Definition 7.11. *Let $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, \ldots, v_n$ are the vertices and $e_1, e_2, \ldots, e_m$ are the edges of G. Then the incidence matrix with* *respect to this ordering of V and E is the $n \times m$ matrix $\mathbf{M} = [m_{ij}]$, where*

$$m_{ij} = \begin{cases} 1 & \textit{when edge } e_j \textit{ is incident with } v_i \\ 0 & \textit{otherwise} \end{cases}$$

Example: Represent the graph shown in Figure below with an incidence matrix.
Solution: The incidence matrix is

$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{cccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

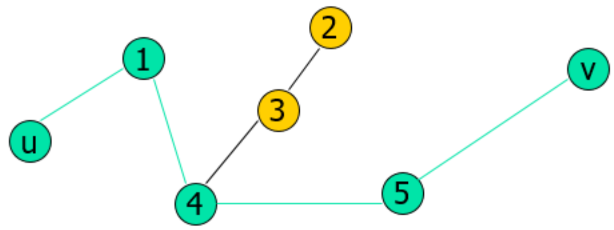
Path and Connectivity

A path is a sequence of edges that begins at a vertex of a graph and travels from vertex to vertex along edges of the graph.

Definition 7.12. *Let n be a nonnegative integer and G an undirected graph. A path of length n from u to v in G is a sequence of n edges $e_1, \ldots, e_n$ of G such that e_1 is associated with $\{x_0, x_1\}$, e_2 is associated with $\{x_1, x_2\}$, and so on, with e_n associated with $\{x_{n-1}, x_n\}$, where $x_0 = u$ and $x_n = v$.*

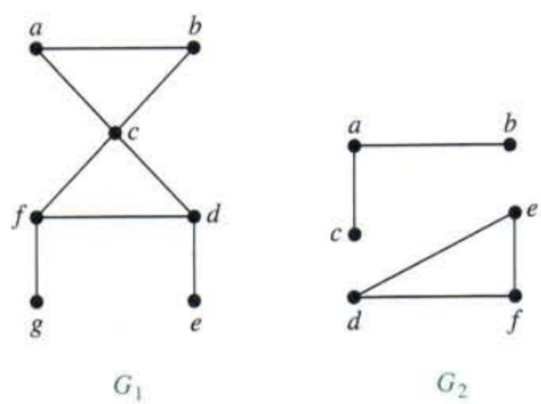
When the graph is simple, we denote this path by its vertex sequence $x_0, x_1, \ldots, x_n$. The path is a $circuit$ if it begins and ends at the same vertex, that is, if $u = v$, and has length greater than zero. The path or circuit is said to pass through the vertices $x_1, x_2, \ldots, x_{n-1}$ or traverse the edges $e_1, e_2, \ldots, e_n$. A path or circuit is simple if it does not contain the same edge more than once.

Representation example: G = (V, E), Path P represented, from u to v is u, 1, 1, 4, 4, 5, 5, v



Definition 7.13. *An undirected graph is called connected if there is a path between every pair of distinct vertices of the graph.*

Example : From the figure, which graph is connected?

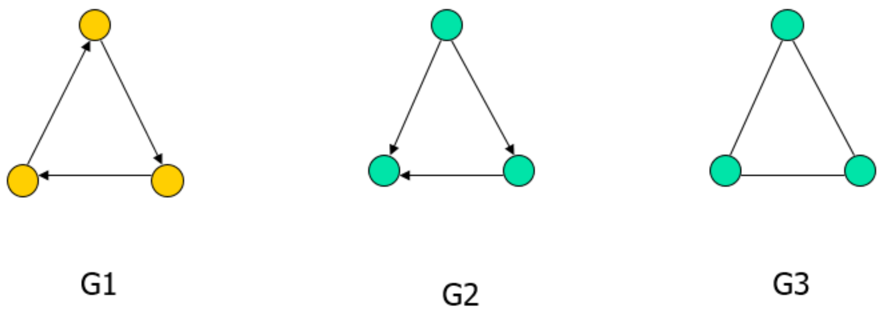


Solution The graph G_1 in Figure is connected, because for every pair of distinct vertices there is a path between them (the reader should verify this). However, the graph G_2 in Figure 2 is not connected. For instance, there is no path in G_2 between vertices a and d.

Theorem 21. *There is a simple path between every pair of distinct vertices of a connected undirected graph.*

Definition 7.14. *A directed graph is strongly connected if there is a path from a to b and from b to a whenever a and b are vertices in the graph.*

Definition 7.15. *A directed graph is weakly connected if there is a path between every two vertices in the underlying undirected graph.*



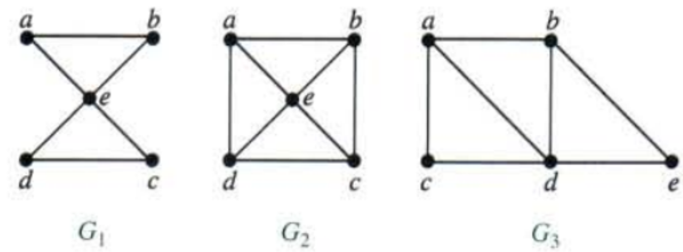
Representation example: G1 (Strong component), G2 (Weak Component), G3 is an undirected graph representation of G2 or G1

7.1.5 Euler and Hamiltonian Path

An Eulerian path (Eulerian trail, Euler walk) in a graph is a path that uses each edge precisely once. If such a path exists, the graph is called traversable.

An Eulerian cycle (Eulerian circuit, Euler tour) in a graph is a cycle that uses each edge precisely once. If such a cycle exists, the graph is called Eulerian (also unicursal).

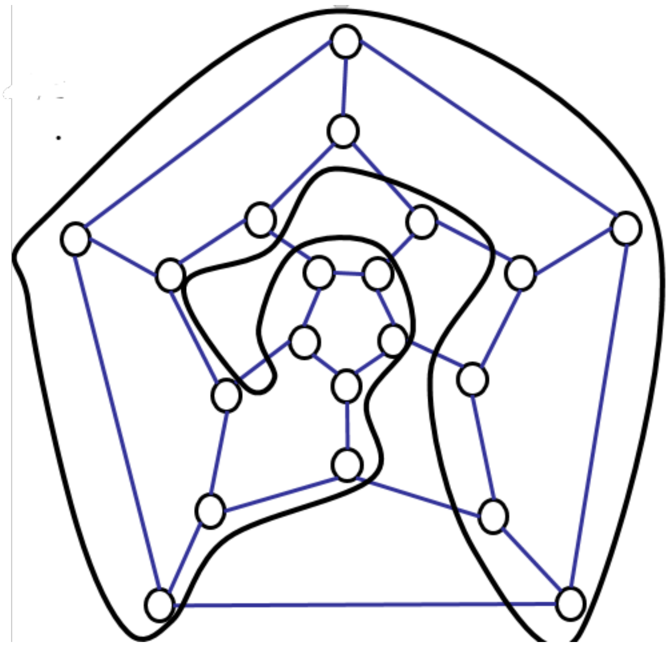
Representation example: from the figures below The graph G_1 has an Euler circuit, for example, a, e, c, d, e, b, a. Neither of the graphs G_2 or G_3 has an Euler circuit (you should verify this). However, G_3 has an Euler path, namely, $a, c, d, e, b, d, a, b. G_2$ does not have an Euler path (you should verify).



Hamiltonian path (also called traceable path) is a path that visits each vertex exactly once.

A Hamiltonian cycle (also called Hamiltonian circuit, vertex tour or graph cycle) is a cycle that visits each vertex exactly once (except for the starting vertex, which is visited once at the start and once again at the end).

A graph that contains a Hamiltonian path is called a traceable graph. A graph that contains a Hamiltonian cycle is called a Hamiltonian graph. Any Hamiltonian cycle can be converted to a Hamiltonian path by removing one of its edges, but a Hamiltonian path can be extended to Hamiltonian cycle only if its endpoints are adjacent.



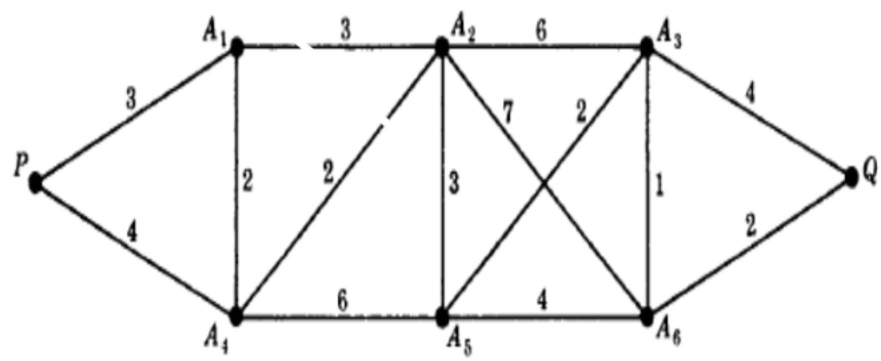
The above graph of the vertices of a dodecahedron is a Hamiltonian .

A graph G is called a $labeled\ graph$ if its edges and/or vertices are assigned data of one kind or another. G is called a $weighted\ graph$ if each edge e of G is assigned a nonnegative number $w(e)$ called the weight or length of v.

The $weight\ (or\ length)\ of\ a\ path$ in such a weighted graph G is defined to be the sum of the weights of the edges in the path.

A $shortest\ path$ is a path of the minimum weight. We can use the Dijkstra Algorithm to find the length of the shortest path between two vertices in a connected simple undirected weighted graph.

Example : Find the length of the shortest path from P to Q.



Solution : The length of the shortest path in the above undirected weighted graph is given by

$$(P, A_1, A_2, A_5, A_3, A_6, Q)$$

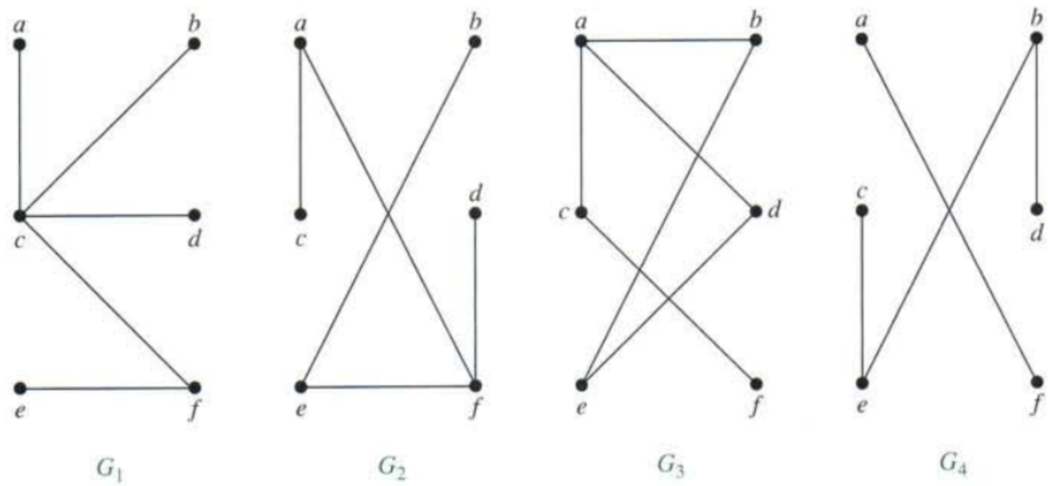
7.1.6 Trees

Trees are particularly useful in computer science, where they are employed in a wide range of algorithms. For instance, trees are used to construct efficient algorithms for locating items in a list. They can be used in algorithms, such as Huffman coding, that construct efficient codes saving costs in data transmission and storage. Trees can be used to study games such as checkers and chess and can help determine winning strategies for playing these games.

Definition 7.16. *A tree is a connected undirected graph with no simple circuits.*

Because a tree cannot have a simple circuit, a tree cannot contain multiple edges or loops. Therefore any tree must be a simple graph.

Example: which of the graphs shown in Figure below are trees?

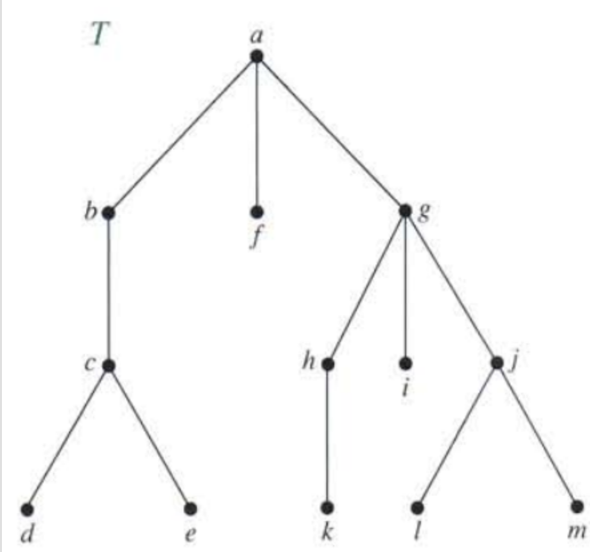


Examples of Trees and Graphs That Are Not Trees.

Solution: G_1 and G_2 are trees, because both are connected graphs with no simple circuits. G_3 is not a tree because e, b, a, d, e is a simple circuit in this graph. Finally, G_4 is not a tree because it is not connected.

Theorem 22. *An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.*

Definition 7.17. *A rooted tree is a tree in which one vertex has been designated as the root and every edge is directed away from the root.*



A Rooted Tree T.

Suppose that T is a rooted tree. If v is a vertex in T other than the root, $the\ parent\ of v$ is the unique vertex $u$such that there is a directed edge from u to v (you should show that such a vertex is unique).

When u is the parent of v, v is called a $child$ of u. Vertices with the same parent are called siblings. The $ancestors$ of a vertex other than the root are the vertices in the path from the root to this vertex, excluding the vertex itself and including the root (that is, its parent, its parent's parent, and so on, until the root is reached).

The $descendants$ of a vertex v are those vertices that have v as an ancestor. A vertex of a tree is called a $leaf$ if it has no children. Vertices that have children are called $internal\ vertices.$ The root is an internal vertex unless it is the only vertex in the graph, in which case it is a leaf.

If a is a vertex in a tree, the $subtree$ with a as its root is the subgraph of the tree consisting of a and its descendants and all edges incident to these descendants.

Example In the rooted tree T (with root a) shown in the above figure , find the parent of c, the children of g, the siblings of h, all ancestors of e, all descendants of b, all internal vertices, and all leaves. What is the subtree rooted at g ?

Solution: The parent of c is b. The children of g are h, i, and j. The siblings of h are i and j. The ancestors of e are c, b, and a. The descendants of b are c, d, and e. The internal vertices are a, b, c, g, h, and j. The leaves are d, e, f, i, k, l, and m. The subtree rooted at g is shown in below.

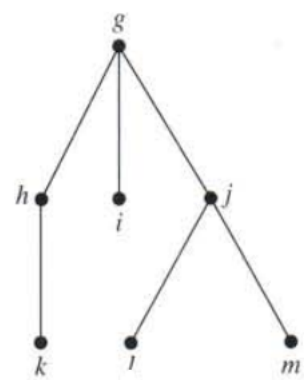
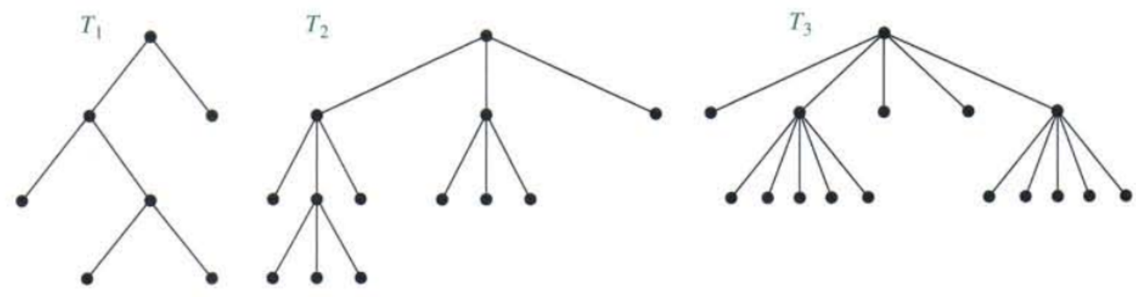


FIGURE The Subtree Rooted at *g*.

Definition 7.18. *A rooted tree is called an m-ary tree if every internal vertex has no more than m children. The tree is called a full m-ary tree if every internal vertex has exactly m children. An m-ary tree with $m = 2$ is called a binary tree.*

Example Are the rooted trees in Figure below full m-ary trees for some positive integer m ?



Solution : T_1 is a full binary tree because each of its internal vertices has two children. T_2 is a full 3-ary tree because each of its internal vertices has three children. In T_3 each internal vertex has five children, so T_3 is a full 5 -ary tree.

7.2 Summary

In this Unit, We have learnt about different types of graphs, graphs which have path or cycles and we have also looked on graphs which are trees.

7.3 End of Unit Test

1. Define a simple graph, a multigraph, a pseudograph, a directed graph, and a directed multigraph.

2. Use an example to show how each of the types of graph in part (a) can be used in modeling. For example, explain how to model different aspects of a compute network or airline routes.

3. What is the relationship between the sum of the degrees of the vertices in an undirected graph and the number of edges in this graph? Explain why this relationship

holds.

4. Why must there be an even number of vertices of odd degree in an undirected graph?

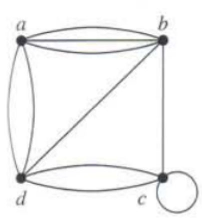
5. What is the relationship between the sum of the in-degrees and the sum of the out-degrees of the vertices in a directed graph? Explain why this relationship holds.

6. Define an Euler circuit and an Euler path in an undirected graph.

7. Describe the famous Königsberg bridge problem and explain how to rephrase it in terms of an Euler circuit.

8. How can it be determined whether an undirected graph

9. Use an adjacency matrix to represent the pseudograph shown in Figure.

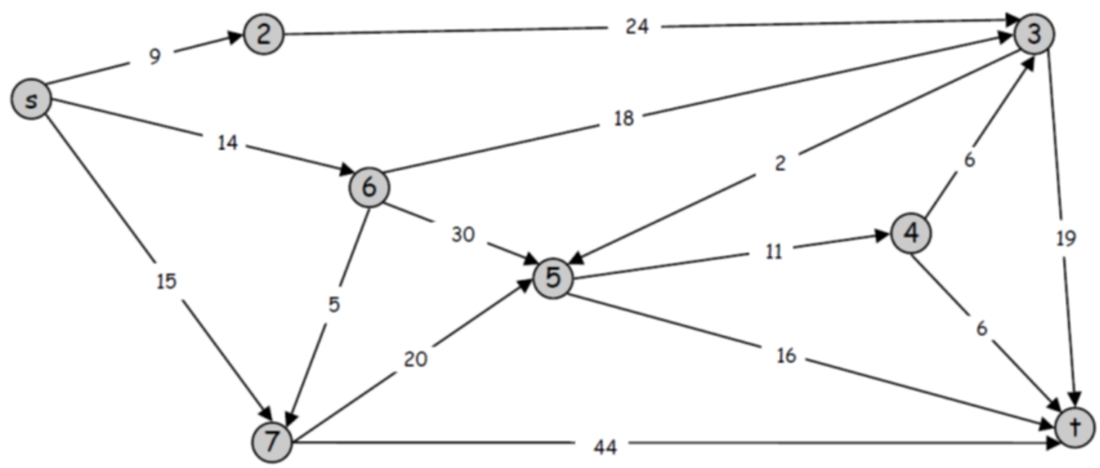


10. Draw an undirected graph represented by the given adjacency matrix.

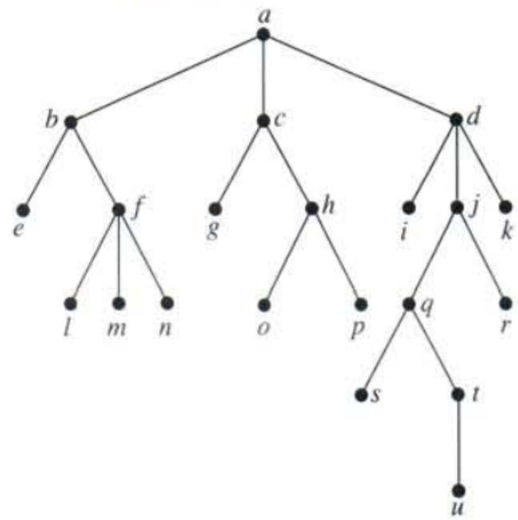
a. $\begin{bmatrix} 1 & 3 & 2 \\ 3 & 0 & 4 \\ 2 & 4 & 0 \end{bmatrix}$

b. $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$

11. Find the shortest path from s to t in the figure below



12. Answer these questions about the rooted tree illustrated.



a) Which vertex is the root?

b) Which vertices are internal?

c) Which vertices are leaves?

d) Which vertices are children of j ?

e) Which vertex is the parent of h ?

f) Which vertices are siblings of o ?

g) Which vertices are ancestors of m ?

h) Which vertices are descendants of b ?

7.4 Further Reading

1. Epp, S.S. (2003). *Discrete Mathematics with Applications*, 4th Edition. Boston: Richard Stratton.

2. Lipschutz, S., Lipson, M. (2005). *Discrete Mathematics*. Tata Mc Graw Hill.

3. Rosen, K.H., (2012). *Discrete Mathematics and its Applications*, New York: McGraw Hill.